



**COMSATS University Islamabad,
Sahiwal Campus, Pakistan**

Health Cloud

**A project presented to
COMSATS Institute Islamabad, Sahiwal Campus**

**In partial fulfillment
of the requirement for the degree of**

Bachelor of Science in Computer Science (2022-2026)

By

Mariyam Rasool	CIIT/SP22-BCS-057/SWL
Muhammad Asif	CIIT/SP22-BCS-067/SWL
Muneeba Faisal	CIIT/SP22-BCS-098/SWL

DECLARATION

We hereby declare that this software, neither whole nor as a part has been copied out from any source. It is further declared that we have developed this software and accompanied report entirely on the basis of our personal efforts. If any part of this project is proved to be copied out from any source or found to be reproduction of some other. We will stand by the consequences. No Portion of the work presented has been submitted of any application for any other degree or qualification of this or any other university or institute of learning.

Mariyam Rasool

Muhammad Asif

Muneeba Faisal

CERTIFICATE OF APPROVAL

It is to certify that the final year project of BS (CS) “**Health Cloud**” was developed by **MARIYAM RASOOL (CIIT/SP22-BCS-057)**, **MUHAMMAD ASIF (CIIT/SP22-BCS-067)** and **MUNEEBA FAISAL (CIIT/SP22-BCS-098)** under the supervision of “**DR. YAWAR ABBAS**” and co supervisor “**DR. MUHAMMAD ALI**” and that in their opinion; it is fully adequate, in scope and quality for the degree of Bachelors of Science in Computer Sciences.

Supervisor

Dr. Yawar Abbas

Lecturer

Department of Computer Science

COMSATS University Islamabad, Sahiwal Campus

External Examiner

Dr. Muhammad Munawar Iqbal

Assistance Professor

Department of Computer Science

UET Taxila

Head of Department

Dr. Zafar Iqbal Roy

Assistant Professor

Department of Computer Science

COMSATS University Islamabad, Sahiwal Campus

Executive Summary

Health Cloud is a centralized healthcare management system that brings patients, doctors, and administrators onto a single digital platform. The system addresses the common problem of fragmented healthcare processes where appointments, medical records, prescriptions, and patient history are stored separately or handled manually. By centralizing these functions, Health Cloud ensures secure storage, easy accessibility, and efficient management of all healthcare information.

The platform includes a web application for doctors, two mobile applications (for patients and doctors), and an admin panel for overall system supervision. It is developed using React TypeScript for the front end, Node.js with Express for the backend, and MongoDB for flexible and secure data storage.

Patients can create accounts, search for doctors by specialty, location, or name, and book appointments digitally. They can also make online payments and store medical records, lab reports, prescriptions, and consultation history. Doctors can access these records only after OTP verification, ensuring privacy and patient consent.

Doctors use a centralized dashboard to view appointments, patient history, consultation details, and generate digital prescriptions. The prescription module is linked with the admin panel, which manages standardized medicine and dosage data. Doctors can also add notes and update patient histories within the same system.

The admin panel supports system integrity by handling doctor approvals, user verification, medicine management, analytics, and overall monitoring.

Overall, Health Cloud modernizes healthcare by integrating essential services into a single cloud-based ecosystem, reducing paperwork, improving coordination, and supporting informed medical decision-making.

Acknowledgement

All praise is to Almighty Allah who bestowed upon us a minute portion of His boundless knowledge by virtue of which we were able to accomplish this challenging task.

We are greatly indebted to our project supervisor “Dr. Yawar Abbas” and our Co-Supervisor “DR. Muhammad Ali”. Without their personal supervision, advice and valuable guidance, completion of this project would have been doubtful. We are deeply indebted to them for their encouragement and continual help during this work.

And we are also thankful to our parents and family who have been a constant source of encouragement for us and brought us the values of honesty & hard work.

Mariyam Rasool

Muhammad Asif

Muneeba Faisal

Abbreviations

Abbreviation	Full Form
API	Application Programming Interface
UI	User Interface
UX	User Experience
EMR	Electronic Medical Records
EHR	Electronic Health Records
HIS	Health Information System
OTP	One-Time Password
DB	Database
SQL	Structured Query Language
NoSQL	Not Only SQL
SDLC	Software Development Life Cycle
MVC	Model-View-Controller
NFR	Non-Functional Requirements
FR	Functional Requirements
CRUD	Create, Read, Update, Delete
JSON	JavaScript Object Notation
HTTPS	Hypertext Transfer Protocol Secure
JWT	JSON Web Token

Table of Contents

1.	Introduction.....	2
1.1	Brief	4
1.1.1	Purpose of the Project	4
1.1.2	Scope of the System.....	4
1.1.3	Technologies Used	5
1.1.4	System Overview	5
1.1.5	Outcome of the Project	5
1.2	Relevance to Course Modules	5
1.2.1	Software Engineering.....	6
1.2.2	Database Systems.....	6
1.2.3	Web Engineering and Internet Programming.....	6
1.2.4	Mobile Application Development.....	7
1.2.5	Object Oriented Programming.....	7
1.2.6	Data Structures and Algorithms	7
1.2.7	Human Computer Interaction	7
1.2.8	Information Security	7
1.2.9	Distributed Computing and Computer Networks	8
1.2.10	Final Year Project I and II	8
1.3	Project Background.....	8
1.4	Literature Review.....	10
1.5	Analysis from Literature Review	11
1.5.1	Alignment with Digital Healthcare Adoption Trends	12
1.5.2	Comparison with Cloud Based Healthcare Research	12
1.5.3	Relevance to Existing Appointment and Patient Management Systems	12
1.5.4	Integration of Electronic Medical Records	13
1.5.5	Application of Security and Privacy Standards	13
1.5.6	Involvement of Mobile Health Concepts	13

1.5.7	Addressing Limitations of Existing Healthcare Technologies.....	13
1.6	Methodology and Software Lifecycle.....	14
1.6.1	Methodology	14
1.6.2	Agile Practices Applied.....	15
1.6.3	Software Development Life Cycle (SDLC) Model	15
1.6.4	Product Backlog Management.....	15
1.6.5	Incremental Delivery	15
1.6.6	Rationale behind Selected Methodology	16
1.6.7	Development Tools and Environment.....	16
1.6.8	Testing and Quality Assurance.....	16
1.6.9	Advantages of the Selected Methodology and SDLC	16
2	Problem Definition.....	18
2.1	Problem Statement	19
2.2	Deliverables and Development Requirements.....	19
2.2.1	Deliverables	19
2.2.1.1	Web Application for Doctors	19
2.2.1.2	Web Application for Doctors	20
2.2.1.3	Mobile Application for Doctors	20
2.2.1.4	Admin Panel.....	20
2.2.1.5	Centralized Cloud Database.....	20
2.2.2	Development Requirements.....	20
2.2.2.1	Technical Requirements.....	20
2.3	Current System.....	21
2.3.1	Manual Record-Keeping.....	22
2.3.2	Fragmented Appointment Management.....	22
2.3.3	Limited Accessibility and Communication.....	23
2.3.4	Security and Privacy Concerns	24
2.3.5	Inadequate Integration of Digital Tools	24
2.3.6	Regional Disparities.....	25

2.3.7	Summary of Limitations	25
2.4	Summary	26
3	Requirement Analysis	29
3.1	Introduction to Requirement Analysis	29
3.2	System Overview	29
3.3	Actors in the System	30
3.3.1	Patient	30
3.3.2	Doctor	31
3.3.3	Staff.....	31
3.3.4	Administrator	31
3.4	Use Case Diagrams	31
3.5	Use Case Descriptions	33
3.5.1	Login/Register Use Case.....	34
3.5.2	Set Profile.....	35
3.5.3	Add Family Member.....	37
3.5.4	Book Appointments	38
3.5.5	Accept/Decline Appointments	39
3.5.6	Access Patient Records	40
3.5.7	Prescribe Medication	41
3.5.8	View Medical Records.....	42
3.5.9	Make Payment	43
3.5.10	Add Family Member.....	44
3.5.11	Approve Doctor Registration.....	45
3.5.12	Manage Finance	46
3.5.13	Manage Users.....	47
3.5.14	Monitor System Logs.....	48
3.5.15	Manage Appointments	49
3.6	Functional Requirements	50
3.6.1	User Management	50

3.6.2	Patient Records Management	51
3.6.3	Appointments Management	51
3.6.4	Prescriptions and Medications	51
3.6.5	Payment and Financial Management	52
3.6.6	Notifications and Alerts	52
3.6.7	Admin and System Management	53
3.6.8	Data Reporting and Analytics	53
3.7	Non-Functional Requirements	53
3.7.1	Performance	54
3.7.2	Scalability	54
3.7.3	Security	54
3.7.4	Reliability and Availability	54
3.7.5	Usability	55
3.7.6	Maintainability	55
3.7.7	Compliance	55
3.7.8	Backup and Recovery	55
3.7.9	Performance Monitoring and Logging.....	56
4	Design And Architecture	58
4.1	System Architecture	58
4.1.1	Architecture Style	58
4.1.2	Components	58
4.1.2.1	User Interface (UI).....	59
4.1.2.2	Authentication Service.....	59
4.1.2.3	Patient Records Module.....	59
4.1.2.4	Appointments Module	59
4.1.2.5	Payment Module	59
4.1.2.6	Admin Dashboard	60
4.1.3	Interactions.....	60
4.1.4	System Architecture Diagram	60

4.2	Data Representation	61
4.2.1	Database Design.....	61
4.2.2	Entity–Relationship Representation.....	62
4.2.3	Data Flow	63
4.2.3.1	Sequence Diagram (Data Flow Across Layers)	64
4.2.3.2	Activity Diagram (Overall System Workflow)	65
4.2.4	Entity Descriptions and Constraints.....	66
4.3	Process Flow / Representation	66
4.3.1	Login and Authentication Process	66
4.3.2	Appointment Booking Process	66
4.3.3	Prescription Generation Process	67
4.3.4	Payment Processing Workflow	67
4.4	Design Models	67
4.4.1	Class Model Description.....	68
4.4.2	Sequence Model Description	68
4.4.3	Activity Model Description	68
4.4.4	Component Model Description.....	68
5	Implementation	70
5.1	Algorithms	70
5.1.1	Authentication Algorithm	70
5.1.2	Appointment Scheduling Algorithm	71
5.1.3	Prescription Generation Algorithm	71
5.1.4	Payment Processing Algorithm.....	72
5.2	External APIs	72
5.3	User Interface.....	74
6	Testing and Evaluation.....	81
6.1	Manual Testing.....	81
6.1.1	System Testing	81
6.1.1.1	Purpose of System Testing.....	82

6.1.1.2	Types of System Testing	82
6.1.1.3	Importance of Manual System Testing	82
6.1.2	Unit Testing.....	82
6.1.3	Functional Testing.....	84
6.1.4	Integration Testing	86
6.2	Automated Testing	87
7	Conclusion	90
7.1	Future Work	90
7.1.1	Online Consultation	90
7.1.2	Hospital Deployment	90
7.1.3	Subscription Models	90
7.1.4	Advanced Analytics and Reporting	91
7.1.5	Integration with Wearable Devices	91
7.1.6	Enhanced Security Features.....	91
8	References.....	92

List of Figures

Figure 3.1: Use case diagram.....	33
Figure 4.1: System Architecture Diagram	61
Figure 4.2: Entity–Relationship Diagram of the System	62
Figure 4.3: Detailed Document Relationship Structure in MongoDB.....	63
Figure 4.4: Layered sequence diagram	64
Figure 4.5: Vertical patient-journey activity diagram	65
Figure 5.1: Home page of doctor app	75
Figure 5.2: Doctor profile	76
Figure 5.3: Doctor home page	77
Figure 5.4: Patient Profile	78
Figure 5.5: Home page web	79

List of Tables

Table 1.1: Comparison of Existing Healthcare Applications and Health Cloud Solution	10
Table 3.1: Structured description of the Login/Register use case.....	34
Table 3.2: Structured description of the Set Profile use case.....	36
Table 3.3: Add Family Member Use Case	37
Table 3.4: Book Appointments Use Case	38
Table 3.5: Accept/Decline Appointments Use Case.....	39
Table 3.6: Access Patient Records Use Case	40
Table 3.7: Prescribe Medication Use Case	41
Table 3.8: View Medical Records Use Case	42
Table 3.9: Make Payment Use Case	43
Table 3.10: Add Family Member Use Case	44
Table 3.11: Approve Doctor Registration Use Case	45
Table 3.12: Manage Finance Use Case	46
Table 3.13: Manage Users Use Case.....	47
Table 3.14: Monitor System Logs Use Case.....	48
Table 3.15: Manage Appointments Use Case	49
Table 5.1: Details of APIs Used in the Project.....	73
Table 6.1: Unit Testing Table	83
Table 6.2: Functional Testing Table	85
Table 6.3: Integration Testing Table.....	86
Table 6.4: Automated Testing Table.....	88

Chapter 1

INTRODUCTION

1. Introduction

Healthcare in Pakistan faces long-standing challenges related to accessibility, coordination, and the secure handling of patient information. A significant portion of healthcare facilities, especially in small cities and rural areas, continue to rely on manual record-keeping methods, handwritten prescriptions, and paper-based appointment systems. As a result, patient histories, laboratory reports, and medical records are often scattered, misplaced, or unavailable when required. These gaps create delays in diagnosis, increase the chances of medical errors, and make it difficult for patients and healthcare providers to maintain continuity of care. With the rapid growth of population, overburdened public hospitals, and limited digital infrastructure in many medical institutions, the need for modern, centralized healthcare solutions in Pakistan has become increasingly urgent.

In recent years, Pakistan has seen a growing shift toward digital services across banking, e-commerce, and transportation. However, the healthcare sector still lacks a unified digital ecosystem that brings patients, doctors, and administrators together. While some private hospitals have introduced electronic systems, they remain isolated within individual institutions and do not offer a nationwide or publicly accessible platform. The absence of centralized digital health records means that when a patient visits different clinics or specialists, their medical data does not follow them. This leads to repeated tests, inconsistent diagnoses, and increased costs for patients. A cloud-based healthcare system has the potential to address these issues by storing medical information centrally and making it accessible through secure digital platforms, regardless of location or device.

Health Cloud is developed as a modern solution to these challenges within the context of Pakistan's healthcare environment. It is a comprehensive digital healthcare management system designed to integrate all major healthcare activities into a single cloud-based platform. The purpose of the system is to digitalize workflows, reduce administrative burdens, improve communication, and make healthcare more accessible to people across Pakistan. Whether a patient is in a large city like Karachi, Lahore, and Islamabad, or in a smaller district, Health Cloud enables them to access healthcare services, maintain medical histories, and communicate with doctors without being limited by physical distance or fragmented record-keeping practices.

The system consists of a doctor web application, mobile applications for patients and doctors, and an admin panel that oversees operations. It is built using modern technologies such as React TypeScript for the frontend and Node.js with Express for the backend, ensuring fast responsiveness, reliability, and scalability. MongoDB is used as the database to store healthcare data, offering flexibility for handling patient profiles, reports, prescriptions, and other structured and unstructured medical information. These technologies support the system's ability to serve a large and diverse user base, which is essential in a country where millions of people require accessible and efficient healthcare solutions.

Health Cloud enables patients in Pakistan to create digital accounts, search for doctors based on specialization, city, consultation fees, and clinic availability, and book appointments online. Since online payments are becoming more common in Pakistan with services such as Easypaisa, JazzCash, and bank gateways, the system allows patients to pay securely and track the status of their appointments. The platform also empowers patients by enabling them to store their medical reports, lab results, prescriptions, and past consultation notes digitally. Through OTP-based verification, patients can allow doctors to access their medical records securely, ensuring privacy and consent-driven data sharing.

Doctors in Pakistan benefit from a dedicated dashboard where they can manage appointments, review patient histories, and generate prescriptions in a standardized format. Many clinics in Pakistan still rely on handwritten prescriptions, which can sometimes be unclear or inconsistent. Health Cloud solves this by providing a structured prescription generation module that is linked to an admin-controlled medicine database. Doctors can also add clinical notes and update patient histories, creating a more complete digital record over time.

The admin panel plays a central role in the ecosystem by verifying doctor credentials, monitoring system activity, managing medicine databases, and ensuring that all information remains authentic and secure. This is particularly important in Pakistan, where the verification of medical practitioners and the prevention of fraudulent activities are significant concerns.

Overall, Health Cloud aims to modernize Pakistan's healthcare system by introducing a cloud-based digital environment where patients, doctors, and administrators can interact seamlessly. The

system reduces dependence on manual methods, minimizes the risks of lost or incomplete medical records, and provides a platform that supports remote access and secure communication. By addressing the unique challenges faced in Pakistan's healthcare sector, Health Cloud represents a practical and scalable solution that has the potential to enhance healthcare accessibility, transparency, and quality across the country.

1.1 Brief

The Health Cloud project is designed as a centralized healthcare management system that connects patients, doctors, and administrators through a unified digital ecosystem. The aim of the project is to overcome the limitations of traditional healthcare systems, where medical data, prescriptions, and appointment details are scattered across different platforms or handled manually. Health Cloud consolidates all essential healthcare services into one platform, ensuring secure storage, easy accessibility, accurate communication, and efficient workflows for all users. The system leverages modern web and mobile technologies to provide a reliable, scalable, and user-friendly healthcare experience.

1.1.1 Purpose of the Project

The primary purpose of Health Cloud is to centralize healthcare operations by offering an all-in-one solution for appointment booking, online payments, medical record management, and digital prescription handling. The project aims to simplify interactions between patients and healthcare providers by removing communication gaps and reducing the need for physical paperwork. It also seeks to empower patients by giving them direct access to their medical history and enabling them to share it securely with authorized doctors whenever needed.

1.1.2 Scope of the System

The scope of Health Cloud extends across multiple platforms, including a web portal for doctors, a mobile application for patients, a mobile application for doctors, and an admin panel for managing system-wide data. The project handles various core functionalities such as doctor discovery based on specialization and location, appointment scheduling, clinical history management, prescription generation, report viewing, and OTP-based secure access to sensitive

patient documents. By covering end-to-end healthcare processes, the system aims to serve as a complete digital solution for clinics, doctors, and patients.

1.1.3 Technologies Used

The Health Cloud system is developed using React TypeScript for the web and mobile interfaces, which provides a modular and scalable structure suitable for large applications. The backend is built using Node.js and Express, offering asynchronous and high-performance server capabilities. MongoDB is used as the primary database because of its flexibility in handling dynamic medical data, patient histories, and large volumes of clinical records. Together, these technologies form a strong foundation for building a cloud-based healthcare platform that can grow over time.

1.1.4 System Overview

Health Cloud operates as an interconnected system that supports users at various levels. Patients can create accounts, search for doctors, book appointments, view prescriptions, upload reports, and manage their medical history. Doctors can access a structured dashboard that displays patient information, appointment requests, uploaded medical reports, and clinical notes. They can also generate prescriptions containing standardized medicines added by the admin panel. The admin panel supervises doctor verification, medicine management, and overall system activity, ensuring smooth functioning and secure data handling across the entire platform.

1.1.5 Outcome of the Project

The project results in a complete healthcare ecosystem that improves the quality and efficiency of patient care. The centralized structure saves time, reduces manual errors, prevents loss of medical records, and enhances communication between patients and doctors. The outcome is a secure, cloud-based digital environment where healthcare information flows seamlessly, enabling doctors to make informed decisions and allowing patients to access trusted medical services without unnecessary delays.

1.2 Relevance to Course Modules

The Health Cloud project is strongly connected with the core subjects taught throughout the Bachelor of Science in Computer Science program. Every stage of the system, from requirement

gathering to implementation and testing, reflects concepts that were learned during the degree. The project demonstrates how theoretical knowledge from various semesters can be applied to develop a complete, centralized healthcare management system that includes a web application, two mobile applications, and an admin panel. The following subheadings explain the contribution of major course modules to the project.

1.2.1 Software Engineering

Software Engineering played a fundamental role in shaping the structure and development of the Health Cloud system. Concepts such as requirement analysis, system modeling, documentation, and SDLC methodologies were applied directly. This project follows structured development practices, including use case modeling, design diagrams, and systematic testing, all of which are core elements covered in the Software Engineering course. The understanding of functional and non-functional requirements, coupled with proper design techniques, allowed the project to be built in an organized and efficient manner.

1.2.2 Database Systems

Database Systems provided the knowledge required to design and implement a secure and flexible database for managing medical records, appointment data, prescriptions, user profiles, and clinic information. The Health Cloud project uses MongoDB, and its schema design reflects core database concepts such as indexing, data retrieval, relationships, normalization principles, and data integrity. Topics such as query optimization, transactions, and database security helped in ensuring that sensitive patient records are stored safely and accessed efficiently.

1.2.3 Web Engineering and Internet Programming

The development of the doctor web application is directly supported by the concepts learned in Web Engineering and Internet Programming. The use of React TypeScript, component based development, responsive layout structures, state handling, API integration, and routing align with the principles introduced in these courses. The courses helped in understanding how to create user friendly, interactive, and dynamic web interfaces that effectively communicate with the backend server.

1.2.4 Mobile Application Development

The mobile applications for patients and doctors are connected to the topics introduced in the Mobile Application Development course. The course provided experience in building application layouts, managing navigation, handling mobile storage, and integrating apps with backend APIs. The concepts of cross platform compatibility, user interface design, and mobile performance optimization were applied to make the applications smooth, responsive, and aligned with real world usage scenarios.

1.2.5 Object Oriented Programming

Object Oriented Programming contributed significantly to organizing the logic and structure of the Health Cloud system. Understanding concepts like classes, objects, inheritance, abstraction, and modularity helped in developing a clean and maintainable codebase. These concepts were used in APIs, business logic layers, and reusable components in both web and mobile interfaces.

1.2.6 Data Structures and Algorithms

Efficient searching, filtering, sorting, and data handling operations within the system are based on the principles learned in Data Structures and Algorithms. Features such as searching for doctors by specialization or location, retrieving medical records, and managing appointment schedules all rely on optimized data operations. These topics ensured that the system performs smoothly even as the number of users grows.

1.2.7 Human Computer Interaction

Human Computer Interaction guided the design of all interfaces in the system. The principles of simplicity, clarity, consistency, and easy navigation were applied to ensure that doctors, patients, and administrators can use the system comfortably. The course helped in creating interfaces that are visually balanced, easy to understand, and suitable for real world healthcare environments.

1.2.8 Information Security

The Health Cloud project handles highly sensitive medical information, so the Information Security course played a major role. Concepts such as authentication, authorization, secure communication, encryption, and data privacy were applied when implementing login systems,

OTP verification, secure access to medical reports, and protection of patient data. The course helped in understanding threats and implementing countermeasures to create a secure system.

1.2.9 Distributed Computing and Computer Networks

The system architecture of Health Cloud relies on server client communication, API integration, and cloud-based data flow, all of which were studied in Distributed Computing and Computer Networks. These courses helped in understanding how different components interact over the network, how requests are processed, and how data travels between the mobile apps, web interfaces, and backend server. Concepts such as latency, packet flow, request handling, and network reliability influenced the overall system design.

1.2.10 Final Year Project I and II

The Final Year Project courses provided the structure, milestones, and academic guidance needed to plan, design, and implement the Health Cloud system. These courses ensured that the project was developed systematically, documented thoroughly, and evaluated through different phases such as proposal, design review, mid evaluation, and final demonstration. The skills gained through these courses shaped the overall discipline and organization of the project.

The Health Cloud project is therefore a complete reflection of the entire BCS degree, demonstrating how multiple subjects combine to form a practical, real world software solution. It shows the integration of programming, design, database management, security, networking, and project management skills that were developed throughout the four-year program.

1.3 Project Background

The idea behind the Health Cloud project originates from the increasing need for a centralized and reliable digital system in the healthcare sector. Traditional healthcare practices rely heavily on manual processes, paper-based records, and disconnected communication channels between patients and doctors. In many clinics, patient files are stored physically, appointments are noted manually, and medical histories are often incomplete or scattered across different places. This creates delays, increases the chances of human error, and makes it difficult for both patients and doctors to access accurate and up to date information when needed.

With the rapid growth of technology and the widespread use of mobile and web applications, healthcare environments require systems that can integrate all essential services into one platform. The concept of a centralized healthcare management system focuses on connecting patients, doctors, and administrators through a unified digital ecosystem. This idea aims to eliminate the gap between medical records, appointments, and prescriptions by allowing all users to interact with the system in a structured and secure way.

The project idea further developed by analyzing the challenges faced by patients in managing their medical information. Many individuals struggle to keep track of previous prescriptions, lab reports, and consultation notes. Doctors also face difficulty in making accurate diagnoses when the patient's history is incomplete or unavailable. By creating a centralized system, Health Cloud ensures that medical records remain organized, secure, and accessible at any time. The system allows doctors to view patient information instantly and enables patients to share their reports with verified doctors through secure OTP based access.

The idea was also influenced by the need for better appointment management. In many clinics, scheduling is manual and often leads to misunderstandings, double bookings, long waiting times, or missed appointments. Health Cloud introduces an automated appointment booking system where patients can view doctors' availability, book a slot, pay online, and wait for confirmation. This creates convenience for patients and helps doctors manage their schedules more efficiently.

Another important factor behind the project idea is the growing demand for digital prescriptions. Handwritten prescriptions can be lost, misunderstood, or misread due to unclear handwriting. A digital prescription system ensures clarity, accuracy, and easy storage. It also enables doctors to prescribe medicines added by the admin panel, ensuring correct dosage and standardization across the platform.

Overall, the idea for Health Cloud is based on creating a modern, cloud-based healthcare ecosystem that addresses the limitations of traditional systems. The project aims to bring convenience, security, and efficiency to the healthcare process by centralizing all essential features into one interconnected platform. It reflects how technology can support better medical care, improve communication, and enhance the patient experience through an organized and reliable digital system.

1.4 Literature Review

The reviewed literature shows that several healthcare applications offer features such as online appointments, doctor discovery, and basic medical record handling. However, each of these platforms also contains important weaknesses that limit their ability to provide a complete and centralized healthcare solution. To clearly highlight these gaps, the following table compares well known healthcare applications with the proposed Health Cloud system. The Table 1.1 presents three key aspects for each application, including the application name, the major weakness identified through literature and practical usage, and the specific solution provided by Health Cloud to overcome that weakness. This comparison helps demonstrate why a unified, cloud-based healthcare platform is necessary and how Health Cloud effectively addresses the limitations found in existing systems.

Table 1.1: Comparison of Existing Healthcare Applications and Health Cloud Solution

Application Name	Weakness	Proposed Project Solution (Health Cloud)
Zocdoc	Focuses mainly on appointment booking and does not provide complete medical record integration.	Health Cloud integrates appointments with full patient histories, prescriptions, and medical reports in one centralized system.
Practo	Limited access controls for sensitive medical documents, and prescriptions are not fully standardized.	Health Cloud includes OTP based secure record access and a standardized prescription generation module connected to an admin-controlled medicine database.
Marham	Does not offer cloud based centralized storage of medical records and lacks structured EMR features.	Health Cloud stores all medical records, reports, and prescription histories on cloud servers with structured EMR formatting.

Application Name	Weakness	Proposed Project Solution (Health Cloud)
Hospital Management Desktop Systems	Installed on local servers, making access limited and lacking multi-device and mobile integration.	Health Cloud provides web and mobile access using cloud technology, allowing record availability from any location.
Manual or Paper Based Clinics	High chances of errors, missing files, and no digital appointment or prescription workflow.	Health Cloud replaces manual work with digital reports, online appointments, automated histories, and secure cloud storage.
Basic EMR Mobile Apps	Do not combine appointment systems, medical records, and prescription modules into one integrated platform.	Health Cloud unifies all features including appointment management, medical histories, clinical notes, and digital prescriptions in one ecosystem.

1.5 Analysis from Literature Review

The analysis of the reviewed literature demonstrates how the Health Cloud project aligns with global trends in digital healthcare transformation and how it specifically addresses the existing gaps within Pakistan's healthcare environment. The literature emphasizes the need for centralized data management, secure medical storage, efficient appointment workflows, and standardized digital prescriptions. These findings support the core features implemented in Health Cloud and validate the relevance of developing such a system within a developing country where digital healthcare adoption is still emerging. The comparison between existing research and the functionalities of Health Cloud shows that the project is not only aligned with modern healthcare solutions but also designed to overcome limitations highlighted in previous studies.

1.5.1 Alignment with Digital Healthcare Adoption Trends

The literature identifies a major shift from paper based healthcare systems to electronic and cloud based platforms worldwide. Research highlights that digital health solutions reduce medical errors, improve access to records, and enhance communication between patients and healthcare providers. The Health Cloud project aligns strongly with these trends by offering a centralized database for storing medical histories, prescriptions, and reports. This alignment shows that the project is built upon well established research and proven benefits of digital healthcare transformation. The system incorporates the same principles discussed in the literature, such as improved accuracy, long term accessibility, and reduced dependency on manual practices.

1.5.2 Comparison with Cloud Based Healthcare Research

Studies on cloud based healthcare solutions highlight advantages such as scalability, remote accessibility, and reduced infrastructure costs [4]. The literature also stresses the importance of secure cloud environments for sensitive medical data. Health Cloud reflects these findings by using cloud technologies and a database structure that supports diverse medical information. The system allows doctors to access patient histories regardless of location, which is consistent with the research on remote healthcare access. The analysis shows that Health Cloud adopts the strengths of cloud computing discussed in the literature and applies them to a practical system that can support patients and doctors across Pakistan.

1.5.3 Relevance to Existing Appointment and Patient Management Systems

Research on platforms like Zocdoc, Practo, and Marham emphasizes the importance of simplifying doctor discovery and appointment booking. These systems mainly focus on connecting patients with doctors, improving scheduling processes, and reducing waiting times. The literature also mentions limitations of these platforms, such as the lack of complete medical record integration. Health Cloud extends beyond the functionality of these existing systems by integrating patient histories, prescriptions, and medical reports within the same platform. Unlike many systems mentioned in the literature, Health Cloud also introduces secure OTP based access, allowing doctors to view sensitive records only with the patient's consent. This comparison indicates that Health Cloud not only meets but expands upon the capabilities of existing solutions.

1.5.4 Integration of Electronic Medical Records

Studies show that electronic medical records improve diagnosis accuracy, reduce duplication of tests, and ensure reliable long term storage. The literature also identifies issues with handwritten prescriptions, such as misinterpretation and inconsistent formatting. Health Cloud directly addresses these concerns through digital record management and standardized prescription generation. The project incorporates every aspect mentioned in the literature, including structured histories, clear medication instructions, and easy accessibility for both patients and doctors. The analysis highlights that Health Cloud builds upon established research but adds an additional layer of centralized accessibility tailored to Pakistan's healthcare context.

1.5.5 Application of Security and Privacy Standards

The literature strongly focuses on security risks associated with healthcare systems and the need for strict authentication, authorization, and encryption mechanisms [1]. Many studies mention that unauthorized access and data breaches are major concerns in digital healthcare environments. Health Cloud implements security mechanisms that match these recommendations, such as OTP verification, role based access, and secure storage of sensitive medical data. The analysis shows that the project is developed in line with internationally recognized healthcare security guidelines, ensuring that the system remains trustworthy and reliable.

1.5.6 Involvement of Mobile Health Concepts

Research on mobile health applications suggests that mobile platforms significantly increase user engagement, improve access to healthcare services, and support better communication between patients and doctors. Health Cloud incorporates these findings by offering a dedicated mobile application for patients and another for doctors. The features included in these apps, such as report viewing, appointment handling, and prescription access, reflect the benefits discussed in mobile health literature. The analysis shows that Health Cloud effectively translates research findings into real world mobile functionalities.

1.5.7 Addressing Limitations of Existing Healthcare Technologies

The literature identifies several shortcomings in current digital healthcare solutions, such as decentralized records, incomplete histories, poor integration between systems, and weak security

measures. Health Cloud is designed specifically to overcome these issues by centralizing data, integrating web and mobile platforms, ensuring secure record sharing, and providing a consistent interface for all users. The comparison indicates that Health Cloud addresses each limitation highlighted in the literature and offers a more complete, unified, and secure healthcare ecosystem.

The analysis shows that Health Cloud is built upon strong foundations supported by research and technological advancements discussed in the literature. The project incorporates the strengths of digital healthcare systems while addressing their weaknesses and adapting them to the needs of Pakistan's healthcare environment. Each reviewed topic, including cloud technologies, medical record management, security practices, online appointments, and mobile health, directly influences the system design and validates the necessity of Health Cloud as a modern solution. The comparison confirms that the project aligns with international healthcare standards while offering features tailored to the challenges faced in Pakistan [5].

1.6 Methodology and Software Lifecycle

The development of the Health Cloud project follows a structured approach that combines modern software engineering practices with a suitable Software Development Life Cycle (SDLC) model. Considering the complexity of the system, which integrates a web application for doctors, mobile applications for patients and doctors, and an admin panel, a methodology that supports iterative development, continuous feedback, and adaptive planning was deemed essential. This approach ensures that all components of the system are thoroughly developed, rigorously tested, and effectively integrated into a cohesive digital healthcare ecosystem. By following a clear methodology, the project team can manage risk, improve productivity, and deliver a robust system that meets functional, non-functional, and security requirements of healthcare services in Pakistan.

1.6.1 Methodology

The Health Cloud project adopts the Agile methodology, which emphasizes iterative development, incremental delivery, and close collaboration among team members. Agile supports continuous reassessment of requirements and frequent delivery of working software, making it ideal for healthcare applications where requirements may evolve due to feedback from medical professionals, regulatory changes, or technological advancements.

1.6.2 Agile Practices Applied

In Health Cloud, Agile practices are implemented through short development cycles called sprints. Each sprint focuses on a specific set of modules, such as patient appointment booking, prescription management, report access, and secure OTP-based verification. Iterative development allows the team to implement, test, and refine features progressively. Feedback from supervisors, healthcare experts, and users is integrated after each sprint, ensuring that the system aligns with real-world healthcare workflows. Agile practices also include daily stand-up meetings, sprint planning, and sprint reviews to monitor progress, address challenges, and plan improvements.

1.6.3 Software Development Life Cycle (SDLC) Model

The project adopts the Scrum model of Agile SDLC due to its suitability for multi-component software development. Scrum divides the project into time-boxed sprints, typically lasting two to four weeks, during which specific tasks are planned, developed, tested, and reviewed. The Scrum framework emphasizes transparency, accountability, and collaboration, ensuring efficient management of tasks across the web, mobile, and admin modules.

1.6.4 Product Backlog Management

Scrum methodology relies heavily on a product backlog, which contains a prioritized list of features, enhancements, bug fixes, and system requirements. In Health Cloud, the backlog includes modules such as doctor verification, appointment scheduling, prescription generation, patient record storage, and online payment processing. The backlog is updated continuously to reflect changing priorities, new requirements, or feedback from stakeholders, ensuring that the development remains aligned with user needs and project objectives.

1.6.5 Incremental Delivery

Incremental delivery is a key principle of Agile and Scrum. Health Cloud is developed in stages, where each sprint delivers a working module that can be tested and evaluated. For example, the first sprint may focus on user registration, login, and profile management, while subsequent sprints implement appointment booking, prescription management, and report sharing. Incremental delivery allows stakeholders to review each component, provide feedback, and ensure that the final system is functional, secure, and user-friendly.

1.6.6 Rationale behind Selected Methodology

The Agile methodology and Scrum SDLC were chosen for Health Cloud because of their adaptability to complex, multi-tier systems. Healthcare applications require frequent updates to comply with changing regulations, incorporate user feedback, and improve usability. Agile provides the necessary flexibility to implement these changes efficiently, while Scrum ensures structured planning, task allocation, and progress tracking. This combination allows for continuous improvement, reduces risk, and maintains high-quality software development throughout the project lifecycle.

1.6.7 Development Tools and Environment

The Health Cloud system is developed using modern technologies such as React TypeScript for frontend web and mobile applications, Node.js with Express for backend services, and MongoDB for database management. These tools were selected to support modular development, scalability, and secure handling of large volumes of sensitive healthcare data. The development environment also includes integrated development tools, version control systems, and testing frameworks that facilitate Agile practices and sprint-based development.

1.6.8 Testing and Quality Assurance

Testing is integrated into every phase of the project to ensure system reliability, security, and functionality. Unit tests, integration tests, and end-to-end testing are performed iteratively during each sprint. Special attention is given to critical healthcare functionalities such as secure access to patient records, OTP verification, prescription accuracy, and appointment scheduling. Quality assurance practices in Health Cloud also include code reviews, peer testing, and validation against functional and non-functional requirements to ensure that the system meets healthcare standards and user expectations.

1.6.9 Advantages of the Selected Methodology and SDLC

The Agile-Scrum approach provides flexibility, adaptability, and continuous improvement, which are crucial for healthcare projects. The methodology ensures early detection of errors, incremental delivery of modules, and ongoing feedback from stakeholders. It promotes collaboration between technical teams and healthcare experts, enhances usability.

Chapter 2

PROBLEM DEFINITION

2 Problem Definition

This chapter discusses the precise problem to be solved by the Health Cloud project. The main issue is the lack of a unified digital platform for managing healthcare operations in Pakistan. Currently, patient information is scattered across multiple locations, appointment scheduling is manual and error-prone, and prescriptions and medical histories are often inconsistent or inaccessible. This fragmented system creates inefficiencies, increases operational costs, compromises patient safety, and hinders timely medical interventions.

Healthcare in Pakistan faces numerous challenges that affect both patients and healthcare providers. The absence of a centralized digital system results in fragmented patient records, delayed diagnoses, inefficient administrative processes, and gaps in communication between doctors and patients. Many clinics and hospitals continue to rely on manual record-keeping, handwritten prescriptions, and paper-based appointment systems, which often lead to lost or incomplete medical histories, repeated tests, and difficulty in maintaining continuity of care. These challenges are particularly pronounced in smaller cities and rural areas, where access to specialized healthcare services and digital infrastructure is limited.

In addition, the growing population and overburdened public hospitals have intensified the need for an organized system that can handle patient data effectively. Patients frequently have to carry physical medical reports from one clinic to another, increasing the risk of misplacement and delayed medical interventions. Doctors struggle to access complete patient histories in a timely manner, which can negatively impact treatment decisions and lead to medical errors. Appointment management is often disorganized, resulting in double bookings, long waiting times, and overall inefficiency in healthcare delivery.

Furthermore, handwritten prescriptions and inconsistent documentation of patient information contribute to medication errors, misdiagnosis, and difficulties in tracking treatment progress. There is also a lack of secure mechanisms for sharing sensitive medical records, which compromises patient privacy and trust. The absence of a centralized system prevents healthcare providers from analyzing patient data efficiently, monitoring treatment outcomes, or implementing evidence-based practices.

2.1 Problem Statement

The healthcare system in Pakistan faces significant challenges due to fragmented patient records, manual appointment scheduling, handwritten prescriptions, and inconsistent data management. Patients often struggle to maintain and share their medical history, leading to repeated tests, delayed diagnoses, and reduced continuity of care. Doctors face difficulties accessing complete patient records, managing appointments efficiently, and generating standardized prescriptions. Administrators lack a unified platform to verify doctors, manage medicines, and monitor clinic operations securely.

The Health Cloud project addresses these issues by providing a centralized, cloud-based healthcare platform. It enables secure storage and retrieval of medical records, OTP-based access to sensitive information, online appointment booking and payments, and standardized prescription management. By integrating patients, doctors, and administrators into a single system, Health Cloud aims to improve efficiency, reduce errors, enhance patient safety, and deliver a reliable, accessible healthcare experience across Pakistan.

2.2 Deliverables and Development Requirements

The Health Cloud project is designed to create a comprehensive digital healthcare management system. This section outlines the main deliverables of the project and the technical and functional requirements necessary for its successful development and deployment. Deliverables define the expected outputs of the project, while development requirements specify the tools, technologies, and system capabilities needed to achieve these outcomes.

2.2.1 Deliverables

The primary deliverables of the Health Cloud project include:

2.2.1.1 Web Application for Doctors

A responsive web application that enables doctors to manage appointments, view patient histories, generate digital prescriptions, and access uploaded medical reports. The web app provides structured dashboards to simplify workflow management and improve efficiency.

2.2.1.2 Web Application for Doctors

A mobile application designed for patients to search for doctors based on specialization, location, and availability. Patients can book appointments, make online payments, upload medical records, view prescriptions, and securely share their health data with authorized doctors through OTP verification.

2.2.1.3 Mobile Application for Doctors

A companion mobile application for doctors to manage appointments, review patient reports, update clinical notes, and generate prescriptions on the go. The app ensures that doctors can provide timely consultations without being tied to the web interface.

2.2.1.4 Admin Panel

An administrative panel that oversees the entire system. The admin panel includes functionalities such as doctor verification, management of medicines and dosage information, monitoring system activities, and ensuring data integrity and security.

2.2.1.5 Centralized Cloud Database

A secure MongoDB database to store patient records, prescriptions, medical reports, appointments, and other healthcare-related information. The database supports scalability, secure access, and efficient retrieval of data.

2.2.2 Development Requirements

The development requirements define the technical needs for building the Health Cloud system.

2.2.2.1 Technical Requirements

The technical requirements define the tools and technologies needed for Health Cloud. React TypeScript is used for responsive web and mobile interfaces. Node.js with Express handles server operations and API development. MongoDB stores medical data securely. Payment gateways are integrated, and cloud hosting ensures scalability and remote access.

- Frontend Development: React TypeScript for responsive and modular web and mobile interfaces.

- Backend Development: Node.js with Express for server-side operations, RESTful API development, and handling business logic.
- Database: MongoDB for secure storage of structured and unstructured medical data.
- Payment Integration: Secure integration with payment gateways popular in Pakistan, such as Easypaisa, JazzCash, and banking APIs.
- Cloud Hosting: Deployment on cloud servers to enable accessibility, scalability, and remote data access.

2.3 Current System

The healthcare system in Pakistan is still largely dependent on traditional, manual, and fragmented processes that limit efficiency, accuracy, and accessibility. Although a few private hospitals and advanced medical centers have begun adopting digital tools, these solutions remain isolated and inconsistent across the sector. There is currently no nationwide or centralized digital platform that combines patient records, doctor–patient communication, appointment scheduling, prescription management, and medical report storage into a single integrated system.

Due to the absence of such a unified platform, both patients and healthcare providers face challenges in maintaining complete medical histories, accessing important health information, and ensuring continuity of care. Many clinics continue to rely on handwritten prescriptions, paper files, and manual appointment registers, which are not only time-consuming but also prone to loss, errors, and duplication of records. Furthermore, digital systems that do exist in some private institutions are restricted to internal use and do not communicate with other clinics or hospitals, resulting in data silos across the healthcare ecosystem.

These gaps become even more evident in semi-urban and rural areas, where digital tools, online systems, and technical infrastructure are either limited or entirely unavailable. Patients often have to travel long distances for consultations while carrying physical documents, and doctors have no reliable way of accessing previous medical records from other facilities. Overall, Pakistan’s current healthcare system lacks the integration, centralization, and digital accessibility required to support modern medical workflows, highlighting the need for a comprehensive solution like Health Cloud.

2.3.1 Manual Record-Keeping

A significant portion of healthcare facilities in Pakistan including small clinics, private practices, rural healthcare centers, and even some hospitals continue to rely heavily on paper-based record-keeping. Patient files, medical histories, prescriptions, diagnostic reports, and billing information are maintained in physical folders. Doctors commonly write prescriptions manually, and laboratories generate printed reports that are stored in patient files or handed directly to the patient.

This manual system creates several operational and clinical challenges. Paper records are vulnerable to human errors such as incomplete documentation, illegible handwriting, and accidental misplacement. In many cases, patients misplace their physical records, forcing healthcare providers to reconstruct medical histories based solely on verbal communication, which may be inaccurate or incomplete. This lack of reliable documentation increases the risk of medication errors, duplicated tests, and delayed clinical decisions [8].

Additionally, the manual filing system imposes a significant administrative burden. Healthcare staff spend considerable time retrieving, organizing, and maintaining physical records, reducing their productivity. Over time, storage space becomes a major constraint, especially in high-volume clinics. These inefficiencies highlight the urgent need for a secure, scalable, digital health record management system that ensures accuracy, accessibility, and continuity of care.

2.3.2 Fragmented Appointment Management

Appointment scheduling in the majority of Pakistani healthcare facilities remains inconsistent, decentralized, and largely manual. Patients typically book appointments through direct phone calls, walk-in registrations, or by coordinating through clinic staff using handwritten appointment books. In some private hospitals, basic standalone scheduling software may be used, but these systems rarely integrate with other departments or external healthcare facilities.

This fragmented approach creates several operational difficulties. Patients often face challenges in finding available time slots, especially when they need to consult multiple specialists. Since there is no centralized platform displaying real-time doctor availability, patients must contact each clinic individually, resulting in inconvenience and time wastage. Healthcare providers also struggle to

manage their schedules effectively, as many still rely on handwritten notes or isolated digital tools that do not synchronize across devices or departments.

Common issues such as double booking, overlapping appointments, inaccurate patient queues, and long waiting times persist due to poor coordination. Clinics often experience bottlenecks, where peak hours become overcrowded and off-peak hours remain underutilized. Missed appointments are also frequent, as patients do not receive automated reminders or notifications.

Ultimately, the absence of a unified appointment management system not only reduces workflow efficiency but also negatively impacts patient satisfaction. Patients may experience frustration due to long queues, uncertainty about waiting times, and difficulty accessing timely care. A centralized, digital appointment management platform would streamline scheduling processes, improve resource utilization, and significantly enhance the patient experience.

2.3.3 Limited Accessibility and Communication

In the existing healthcare system, the accessibility of patient information is highly restricted due to the absence of centralized, digitalized record-sharing mechanisms. Patient data is typically stored within the premises of a single clinic, hospital, or laboratory, making it nearly impossible for other healthcare providers to retrieve or review the patient's medical history when needed. This becomes particularly problematic when patients seek second opinions, change healthcare providers, or consult specialists in different cities or regions. As a result, continuity of care is disrupted, and physicians often have to rely on incomplete or verbally communicated medical information provided by patients.

Communication channels between doctors and patients are equally limited and inefficient. Most interactions occur through physical visits, phone calls, or manual messages managed by clinic staff. These communication methods often lead to missed notifications regarding appointment changes, delays in receiving lab results, or unavailability of timely updates about treatment plans. Important reminders such as follow-up appointments, medication schedules, or precautionary guidelines are seldom automated, leading to higher chances of patient non-compliance.

2.3.4 Security and Privacy Concerns

The current healthcare infrastructure does not adequately safeguard sensitive patient information. Most clinics rely on physical files, unprotected standalone computers, or basic local software to store medical records. These systems are vulnerable to a range of threats, including accidental loss, physical damage from environmental conditions, or unauthorized access by untrained personnel. Due to the absence of standardized digital security protocols, confidential records may be viewed, altered, or misplaced without detection.

Privacy concerns are particularly significant because medical information is highly personal and sensitive [7]. Patients may be reluctant to share their full medical details due to fear of exposure or mishandling, which can lead to incomplete diagnoses or inaccurate treatment plans. Weak security practices not only violate ethical and legal obligations but also reduce patient trust in the healthcare system. These gaps are more pronounced in small private clinics where oversight mechanisms, compliance standards, and digital literacy levels are often low. Ensuring robust data security is therefore essential for protecting patient confidentiality and enhancing the credibility of healthcare providers.

2.3.5 Inadequate Integration of Digital Tools

Although some private hospitals in Pakistan have introduced digital systems for billing, patient management, or appointment scheduling, these tools are typically isolated and designed to operate within a single institution. There is no unified platform or national-level infrastructure to interconnect healthcare facilities, resulting in fragmented digital ecosystems. Doctors working in different clinics cannot access shared patient records or standardized medical histories, and medical documents generated by one system are often incompatible with those used in another.

The lack of integration prevents the smooth flow of information across departments such as admissions, pharmacy, laboratory, and radiology leading to repetitive tests, duplicated paperwork, and increased operational inefficiencies. For example, a patient's diagnostic report generated at one laboratory cannot be automatically shared with a physician in another clinic, forcing the patient to deliver physical documents manually.

Overall, the inadequate integration of digital tools significantly limits the potential of technology to enhance healthcare services. A centralized, interconnected system is essential for ensuring that patient data flows seamlessly across facilities, enabling more accurate diagnoses, improved coordination, and better long-term health outcomes.

2.3.6 Regional Disparities

Regional disparities further intensify the challenges within Pakistan’s healthcare system, creating a noticeable divide between urban centers and rural or remote regions. While major cities may have partial digital setups or isolated hospital information systems, smaller towns and villages often lack even the most basic technological infrastructure. Many healthcare facilities in these areas operate without computers, reliable internet access, or trained personnel to manage digital workflows. As a result, patient records are maintained entirely on paper, and appointment scheduling is conducted verbally or through handwritten registers. The absence of digital tools means that even simple tasks such as retrieving a past prescription, confirming a diagnosis, or reviewing medical history become difficult and time-consuming.

Patients in rural regions also face significant travel burdens, often journeying long distances to consult a specialist or access diagnostic services not available locally. This not only increases healthcare costs but also causes delays in receiving timely treatment. Moreover, when rural patients are referred to hospitals in larger cities, the new healthcare providers have no access to previous medical records, forcing them to rely solely on the patient’s memory or incomplete documentation. This results in fragmented care, repeated diagnostic tests, and preventable delays in treatment. These disparities create unequal healthcare experiences, contributing to poorer health outcomes in underdeveloped regions. The lack of a connected digital ecosystem therefore widens the urban–rural healthcare gap and highlights the pressing need for a centralized and universally accessible digital healthcare platform.

2.3.7 Summary of Limitations

In summary, Pakistan’s current healthcare system is hindered by multiple structural and operational limitations that collectively compromise efficiency, accuracy, and patient satisfaction. The heavy dependence on manual processes leads to fragmented patient records, inconsistent data

storage, and frequent loss or misplacement of important medical documents. Doctors often lack complete patient histories, which affects the quality of diagnosis and treatment decisions, especially in emergency situations. Appointment management remains equally inefficient, as most clinics rely on handwritten schedules or local registers that cannot synchronize across multiple facilities. This results in long waiting times, scheduling conflicts, and poor coordination between healthcare providers.

These issues are amplified in rural and underdeveloped regions, where limited technological resources create further barriers to efficient healthcare delivery. Collectively, these limitations demonstrate the urgent need for a secure, centralized, cloud-based healthcare management system. A system like Health Cloud would address these challenges by providing uniform access to patient records, improving the accuracy and speed of clinical decision-making, enabling seamless communication, enhancing data security, and reducing disparities across regions. By digitizing and integrating healthcare processes, Health Cloud has the potential to significantly elevate the overall quality, accessibility, and reliability of healthcare services in Pakistan.

2.4 Summary

The second chapter of this report presents a detailed exploration of the challenges within Pakistan's current healthcare system and establishes the necessity for a unified, digital solution such as Health Cloud. The chapter begins by defining the core problem Pakistan lacks a centralized, cloud-based healthcare platform capable of integrating patient records, appointment management, prescriptions, medical reports, secure communication, and administrative workflows. This fragmented environment creates systemic inefficiencies that negatively impact patients, doctors, and healthcare administrators alike.

The chapter highlights how the absence of digital integration results in scattered patient histories, delayed diagnoses, inconsistent documentation, and poor continuity of care. Many healthcare facilities still rely on handwritten prescriptions, paper files, manual appointment registers, and isolated digital tools that operate independently. As a result, patient information becomes difficult to track, easy to misplace, and almost impossible to access across different clinics or hospitals.

These limitations are further intensified by Pakistan's rapidly growing population, increased pressure on public hospitals, and the rising need for accurate, timely, and accessible clinical data.

The chapter concludes that these challenges manual processes, fragmented systems, limited data accessibility, weak communication, security vulnerabilities, and regional inequalities collectively demonstrate the urgent need for a centralized, integrated healthcare management system. Health Cloud is presented as a comprehensive solution designed to digitize and harmonize healthcare operations across Pakistan. By offering secure cloud storage, integrated appointment systems, digital prescriptions, standardized workflows, and multi-platform accessibility, Health Cloud aims to enhance healthcare efficiency, improve patient safety, strengthen administrative oversight, and ensure continuity of care. Through this centralized ecosystem, the project strives to transform Pakistan's healthcare landscape from a fragmented system into a unified, modern, and accessible digital framework.

Chapter 3

REQUIREMENT ANALYSIS

3 Requirement Analysis

The Requirement Analysis chapter provides a comprehensive and structured understanding of what the proposed system must accomplish in order to function effectively within the intended healthcare environment. It serves as a foundation for translating user expectations into clear technical requirements by identifying system behavior, user interactions, and the specific tasks the system must perform. This chapter outlines how the Health Cloud system is expected to operate, detailing both functional and non-functional requirements while defining the roles and responsibilities of each actor involved, such as patients, doctors, staff, and administrators. Using the provided use case diagram as a reference, this chapter organizes the requirements into well-defined sections that guide the design, development, and validation processes. Through this analysis, the development team gains a precise understanding of the system's objectives, constraints, and performance expectations, ensuring that the final implementation aligns with the needs of all stakeholders and effectively addresses the limitations of the current healthcare system.

3.1 Introduction to Requirement Analysis

Requirement Analysis is a critical phase in the software development lifecycle, serving as the foundation upon which the entire system is designed, implemented, and evaluated. For the Health Cloud system, this phase ensures that all functional expectations, performance needs, and operational constraints are clearly understood before development begins. The purpose of conducting requirement analysis is to bridge the gap between user needs and technical implementation, ensuring the final system effectively resolves existing healthcare challenges such as fragmented processes, manual record handling, and inefficient communication across medical facilities. This chapter outlines the detailed requirements of the Health Cloud platform, describing how each system component interacts with users, how the data flows, and what outcomes are expected from every interaction.

3.2 System Overview

The Health Cloud system is designed as an integrated digital platform that connects patients, doctors, administrative staff, and healthcare facilities within a unified environment. The system enables patients to manage appointments, access medical histories, and communicate with

healthcare providers. Doctors, on the other hand, can view patient records, write prescriptions, manage appointments, and maintain updated case histories. Hospital staff play an essential supporting role by verifying registrations, maintaining patient queues, and assisting with daily workflow operations. Meanwhile, the administrator oversees the entire system by managing user accounts, monitoring system performance, and ensuring that policies, security standards, and database integrity are maintained.

The use case diagram illustrates the relationships between these actors and the system, showing how functionalities such as booking appointments, generating prescriptions, maintaining patient profiles, and managing doctor schedules are interconnected. This overview reflects the system's role as a centralized hub that replaces manual procedures with a streamlined, cloud-based solution accessible from any location. By consolidating all these processes, the Health Cloud platform addresses key issues of data fragmentation, inconsistent communication, and lack of uniformity in patient care.

3.3 Actors in the System

The Health Cloud system interacts with multiple categories of users, each performing distinct roles that contribute to the overall functionality of the platform. These users, also known as actors, engage with the system through specific use cases that reflect their responsibilities, permissions, and operational needs. Understanding these actors and their interactions is crucial for defining accurate system requirements and ensuring that the platform delivers a seamless and secure experience. The following sections describe each actor in detail, outlining their characteristics, goals, and the actions they perform within the Health Cloud ecosystem.

3.3.1 Patient

The patient is the primary beneficiary of the Health Cloud system. Patients interact with the platform to register themselves, book or reschedule appointments, and maintain access to their medical history. Through the system, patients can also receive prescriptions, notifications, and follow-up schedules directly from the doctor. The system ensures that patients can conveniently manage their healthcare journey without relying on physical documents or repeated hospital visits.

3.3.2 Doctor

Doctors are essential users who utilize the platform to review patient histories, confirm appointments, write prescriptions, and maintain consultation records. The system enables doctors to access real-time information, ensuring accuracy in clinical decision-making. The digital interface helps doctors reduce time spent on paperwork and enhances the quality of care by providing organized, easily accessible patient information.

3.3.3 Staff

Hospital staff interact with the system to manage patient queues, verify appointments, and maintain smooth workflow operations. Their role ensures that patient traffic is handled efficiently and that all registrations, updates, and verifications adhere to hospital protocols. The system supports staff in minimizing manual labor, reducing administrative delays, and keeping service delivery organized.

3.3.4 Administrator

The administrator oversees the entire Health Cloud platform. Their responsibilities include account management, system configuration, policy implementation, and ensuring data security. Administrators handle the approval of doctor and staff accounts, manage system logs, and ensure that the database remains consistent and secure. Their control panel provides tools for monitoring system health and generating reports for operational improvements.

3.4 Use Case Diagrams

Use case diagrams provide a clear and structured representation of how different users interact with the Health Cloud system. They help define the system's functional boundaries and illustrate the various operations performed by each actor. Figure 3.1 shows the primary use case diagram for the Health Cloud platform, based on the requirements identified for the system.

In Figure 3.1, four main actors Patient, Doctor, Staff, and Admin interact with the system through a set of defined use cases. The Patient uses the system to register an account, log in, search for doctors, book appointments, upload medical reports, and access prescriptions. The Doctor interacts with the system by managing appointments, reviewing patient histories, generating digital prescriptions, and updating medical records. The Staff actor is responsible for verifying

appointments, maintaining internal workflows, and assisting in daily operational tasks. The admin oversees system-level functionalities such as managing user accounts, monitoring platform activity, and ensuring security and compliance.

The diagram also highlights how these actors connect with key system functionalities, such as appointment scheduling, medical history management, report uploads, and secure communication. Each use case represents a specific function that contributes to the overall healthcare workflow. By visualizing these interactions, the use case diagram ensures that all user needs, functional requirements, and system behaviors are correctly captured.

Overall, Figure 3.1 serves as a foundational model for understanding the operational structure of the Health Cloud system. It guides developers and stakeholders by clearly depicting what the system is expected to perform and how different users will interact with it. This helps eliminate ambiguity, supports accurate requirement validation, and ensures that the system design aligns with the intended objectives of the project.

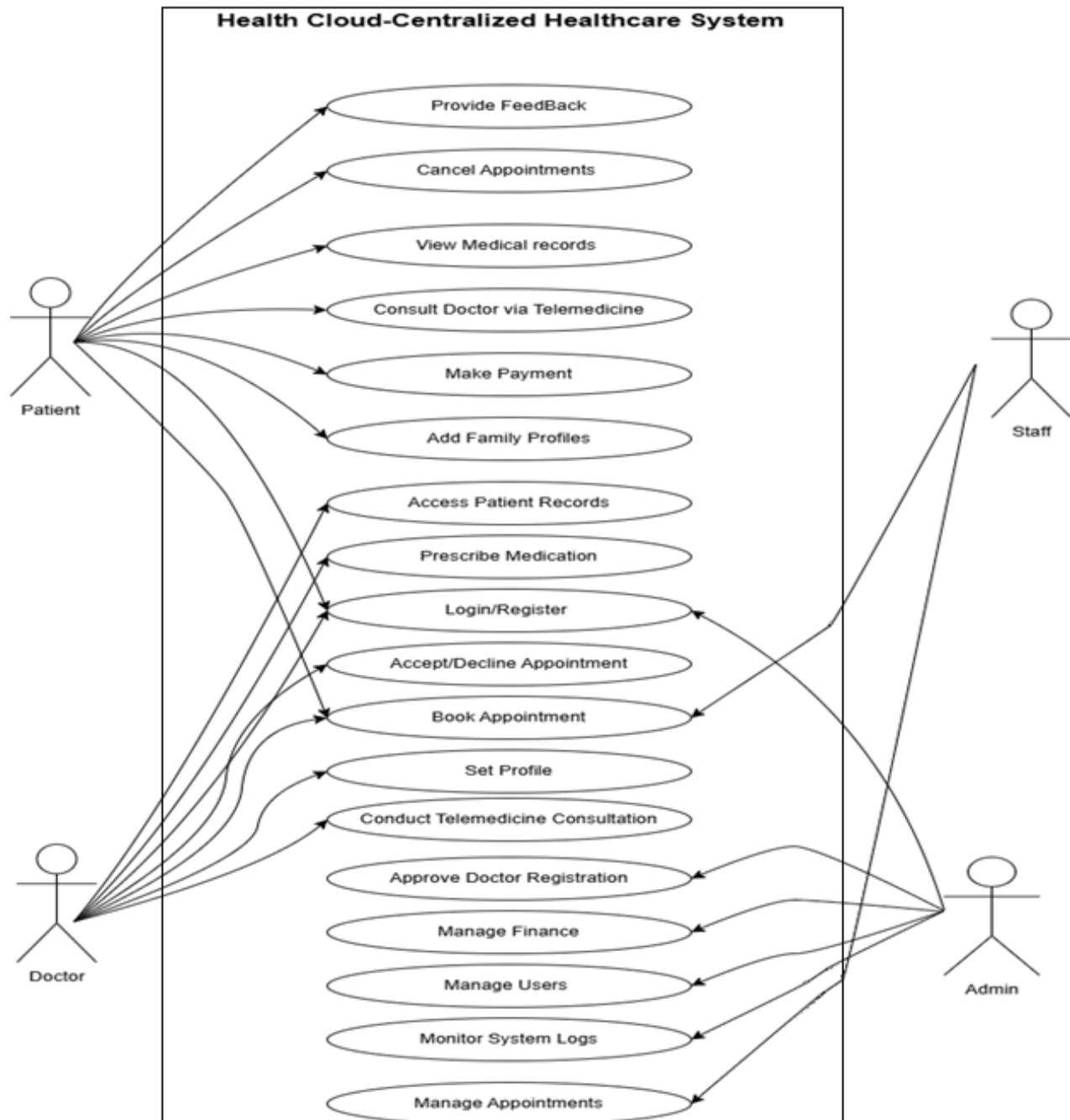


Figure 3.1: Use case diagram

3.5 Use Case Descriptions

This section provides detailed descriptions of all identified use cases. Each use case outlines actors involved, triggers, preconditions, postconditions, normal and alternative flows, exceptions, and

business rules. These tables collectively define the full system behavior and serve as the foundation for design and development.

The following tables present the detailed descriptions of the 17 major use cases included in the Health Cloud system. These use cases highlight how users interact with the system to perform essential healthcare tasks such as booking appointments, viewing medical records, managing finances, conducting telemedicine consultations, and monitoring system activities. Each table follows a standardized template to ensure clarity and consistency in requirement documentation.

3.5.1 Login/Register Use Case

The Login/Register use case defines how different users Patients, Doctors, and Administrators gain access to the Health Cloud system. This Table 3.1 use case ensures that every user is properly authenticated before interacting with the system's features. It covers both new user registration and secure login for returning users. The process includes entering credentials, validating information, and granting system access based on user roles. This use case is essential for maintaining security, preventing unauthorized access, and ensuring that each user interacts with the system according to their designated permissions.

Table 3.1: Structured description of the Login/Register use case.

Use Case Attribute	Details
Use Case Name	User Login & Registration
Primary Actors	Patient, Doctor, Admin
Secondary Actors	Authentication Service, Database
Description	This use case enables users to create an account or log in to the Health Cloud system. The system verifies user credentials, validates identities, and grants access based on their assigned roles.
Trigger	A new user attempts to register, or an existing user attempts to log in.

Pre-Conditions	• User has a stable internet connection. • User must have a valid email/phone number for verification. • Doctor and Admin accounts must be pre-approved by the system (for role verification).
Post-Conditions	• User is successfully authenticated. • User is redirected to their respective dashboard (Patient, Doctor, Admin).
Normal Flow	1. User selects “Login” or “Register.” 2. User enters required information. 3. System verifies credentials/OTP. 4. System check’s role and authentication status. 5. Access is granted and user is logged in.
Alternative Flow	• Invalid username/password entered → System displays an error and prompts retry. • Incorrect OTP for registration → System requests re-verification.
Exceptions	• Server downtime during authentication. • No internet connection. • Database unreachable.
Business Rules	• Authentication must comply with secure login standards (hashed passwords, OTP verification). • Doctors and Admins must be verified before account activation. • Multiple failed attempts may trigger temporary account lock for security.

3.5.2 Set Profile

The Table 3.2 “Set Profile” use case focuses on allowing doctors to manage and update their profiles within the Health Cloud system. Doctors can modify details such as their specializations,

professional experience, and availability. This functionality is triggered whenever a doctor wants to update their profile. The system ensures that the doctor is logged in before allowing any changes. Normally, the doctor logs in, navigates to the profile section, edits the necessary details, and saves the changes, resulting in a successful update. Alternative flows handle cases where updates fail due to invalid input, and exceptions include server errors or loss of internet connectivity. Business rules ensure that profiles are verified before being displayed publicly, maintaining the accuracy and reliability of the information.

Table 3.2: Structured description of the Set Profile use case.

Use Case Attribute	Details
Use Case Name	Set Profile
Primary Actors	Doctor
Description	Doctors can update their profiles with specializations, experience, and availability.
Trigger	Doctors want to update their profile.
Pre-Conditions	Doctor is logged in.
Post-Conditions	Profile is updated successfully.
Normal Flow	1. Login → 2. Navigate to profile → 3. Edit details → 4. Save changes
Alternative Flow	Profile update fails due to invalid input
Exceptions	Server error, no internet
Business Rules	Profiles must be verified before appearing publicly

3.5.3 Add Family Member

The Table 3.3 “Add Family Member” use case allows patients to add family members to their Health Cloud account for shared health tracking. This functionality is triggered whenever a patient wants to include a dependent under their account. The system requires that the patient is logged in before performing this action. Normally, the patient logs in, navigates to the profile section, adds the family member’s details, and confirms the addition, resulting in a successful update. Alternative flows account for situations where invalid details are provided, while exceptions include server errors, lack of internet connectivity, or duplicate entries. Business rules enforce that each patient can add only a limited number of dependents, ensuring proper management and data integrity.

Table 3.3: Add Family Member Use Case

Use Case Attribute	Details
Use Case Name	Add Family Member
Primary Actors	Patient
Description	The patient can add family members to their account for shared health tracking.
Trigger	The patient wants to add a dependent.
Pre-Conditions	Patient must be logged in.
Post-Conditions	Family members are added successfully.
Normal Flow	1. Patient logs in → 2. Navigates to profile → 3. Adds family details → 4. Confirms addition
Alternative Flow	Invalid details provided
Exceptions	Server error, no internet, or duplicate entries

Business Rules	Each patient can add a limited number of dependents
----------------	---

3.5.4 Book Appointments

The “Book Appointments” use case enables patients, doctors, or staff to schedule appointments within the Health Cloud system. This use case is triggered whenever a patient or staff member attempts to book a session with a doctor. As outlined in Table 3.4, the pre-condition requires that the doctor has available slots before an appointment can be scheduled. The normal flow involves the user logging in, selecting a doctor and an available time slot, and confirming the booking, which results in a successfully scheduled appointment. Alternative flows handle cases where the chosen slot is unavailable, and exceptions include server errors or loss of internet connectivity. Business rules ensure that overbooking is strictly prohibited to maintain proper scheduling and resource management.

Table 3.4: Book Appointments Use Case

Use Case Attribute	Details
Use Case Name	Book Appointments
Primary Actors	Patient, Doctor, Staff
Description	Patients, doctors, or staff can schedule an appointment.
Trigger	A patient or staff member books a session.
Pre-Conditions	The doctor has available slots.
Post-Conditions	The appointment is scheduled successfully.
Normal Flow	1. Login → 2. Select doctor and time slot → 3. Confirm booking
Alternative Flow	Slot is unavailable

Exceptions	Server error, no internet
Business Rules	Overbooking is not allowed

3.5.5 Accept/Decline Appointments

The “Accept/Decline Appointments” use case allows doctors to manage appointment requests from patients within the Health Cloud system. As shown in Table 3.5, this use case is triggered whenever a patient books an appointment. The pre-condition requires that the doctor has available slots to accommodate appointments. In the normal flow, the doctor logs in, navigates to the appointment requests, and either accepts or declines each request, with the patient being notified of the outcome. Alternative flows handle scenarios where appointments are automatically declined if the doctor does not respond within a specified time limit. Exceptions include system downtime or doctor unavailability. Business rules mandate that doctors respond to appointment requests within a set timeframe to ensure proper scheduling and patient satisfaction.

Table 3.5: Accept/Decline Appointments Use Case

Use Case Attribute	Details
Use Case Name	Accept/Decline Appointments
Primary Actors	Doctor
Description	Doctors can accept or decline appointment requests from patients.
Trigger	A patient booked an appointment.
Pre-Conditions	The doctor has available slots.
Post-Conditions	The appointment is accepted or declined, and the patient is notified.

Normal Flow	1. Doctor logs in → 2. Navigates appointment requests → 3. Accepts or declines
Alternative Flow	Appointment is auto declined if not responded to within a time limit
Exceptions	System downtime, doctor unavailability
Business Rules	Doctors must respond within a set time frame

3.5.6 Access Patient Records

The “Access Patient Records” use case allows both patients and doctors to view health records within the Health Cloud system. As illustrated in Table 3.6, this use case is triggered whenever a doctor needs to review a patient’s medical history or a patient wants to check their own data. The pre-condition requires that the patient has medical records stored in the system. Normally, the user logs in, navigates to the records section, and views the history. Alternative flows handle cases where no records are available, while exceptions include data retrieval errors, lack of internet connectivity, or unauthorized access. Business rules ensure that only authorized users are allowed to access patient records, maintaining privacy and data security.

Table 3.6: Access Patient Records Use Case

Use Case Attribute	Details
Use Case Name	Access Patient Records
Primary Actors	Patient, Doctor
Description	Both patients and doctors can access health records.
Trigger	A doctor needs patient history, or a patient checks their data.
Pre-Conditions	Patient must have medical records in the system.
Post-Conditions	Requested data is successfully accessed.

Normal Flow	1. Login → 2. Navigate to records → 3. View history
Alternative Flow	No records available
Exceptions	Data retrieval error, no internet, or unauthorized access
Business Rules	Only authorized users can access records

3.5.7 Prescribe Medication

The “Prescribe Medication” use case enables doctors to issue prescriptions and patients to view them within the Health Cloud system. As shown in Table 3.7, this use case is triggered whenever a doctor introduces a new prescription. The pre-condition requires that the patient has a valid consultation. In the normal flow, the doctor logs in, accesses the patient’s data, and issues the prescription, which is then saved in the system. Alternative flows account for situations where prescription approval is required, and exceptions include system errors, lack of internet connectivity, or unauthorized access. Business rules ensure that only licensed doctors are authorized to prescribe medication, guaranteeing legal compliance and patient safety.

Table 3.7: Prescribe Medication Use Case

Use Case Attribute	Details
Use Case Name	Prescribe Medication
Primary Actors	Patient, Doctor
Description	The doctor can prescribe medication, and the patient can view prescriptions.
Trigger	A doctor is introducing a new prescription.
Pre-Conditions	Patient has a valid consultation.
Post-Conditions	Prescription is saved in the system.

Normal Flow	1. Doctor logs in → 2. Accesses patient data → 3. Issues prescription
Alternative Flow	Prescription approval required
Exceptions	System error, no internet, or unauthorized access
Business Rules	Only licensed doctors can issue prescriptions

3.5.8 View Medical Records

The “View Medical Records” use case allows patients to access their past medical history, prescriptions, and test reports within the Health Cloud system. As shown in Table 3.8, this use case is triggered whenever a patient wants to check their medical history. The pre-condition requires that the patient is logged in. In the normal flow, the patient logs in, navigates to the medical records section, and views the details. Alternative flows account for situations where the patient has no records or the system is under maintenance, while exceptions include no internet connection, server downtime, or data retrieval errors. Business rules ensure that only the patient and authorized doctors can view the records, maintaining privacy and security.

Table 3.8: View Medical Records Use Case

Use Case Attribute	Details
Use Case Name	View Medical Records
Primary Actors	Patient
Description	The patient can view their past medical history, prescriptions, and test reports.
Trigger	The patient wants to check their medical history.
Pre-Conditions	Patient must be logged in.
Post-Conditions	Patient successfully views medical records.

Normal Flow	1. Patient logs in → 2. Navigates to medical records → 3. Views details
Alternative Flow	Patient has no records, or the system is under maintenance
Exceptions	No internet connection, server down, or data retrieval error
Business Rules	Only the patient and authorized doctors can view records

3.5.9 Make Payment

The “Make Payment” use case enables patients to pay for consultations, prescriptions, or other services within the Health Cloud system. As illustrated in Table 3.9, this use case is triggered whenever a patient wants to complete a payment. The pre-condition requires that the patient has an active session or a pending bill. Normally, the patient logs in, navigates to the billing section, selects a payment method, and completes the transaction. Alternative flows handle cases where payment fails due to insufficient funds or incorrect details. Exceptions include payment gateway errors, lack of internet connectivity, or bank server issues. Business rules ensure that all transactions are secure to maintain confidentiality and financial safety.

Table 3.9: Make Payment Use Case

Use Case Attribute	Details
Use Case Name	Make Payment
Primary Actors	Patient
Description	The patient can make payments for consultations, prescriptions, or services.
Trigger	The patient wants to pay for services.
Pre-Conditions	Patient has an active session or pending bill.
Post-Conditions	Payment is successfully processed.

Normal Flow	1. Patient logs in → 2. Navigates to billing → 3. Selects payment method → 4. Completes transaction
Alternative Flow	Payment fails due to insufficient funds or incorrect details
Exceptions	Payment gateway error, no internet, or bank server issues
Business Rules	Secure transactions must be ensured

3.5.10 Add Family Member

The “Add Family Member” use case allows patients to add family members to their account for shared health tracking. As shown in Table 3.10, this use case is triggered whenever a patient wants to include a dependent in their account. The pre-condition requires that the patient is logged in. Normally, the patient logs in, navigates to the profile section, adds the family member’s details, and confirms the addition. Alternative flows handle invalid details, while exceptions include server errors, no internet connectivity, or duplicate entries. Business rules ensure that each patient can add only a limited number of dependents to maintain proper account management and data integrity.

Table 3.10: Add Family Member Use Case

Use Case Attribute	Details
Use Case Name	Add Family Member
Primary Actors	Patient
Description	The patient can add family members to their account for shared health tracking.
Trigger	The patient wants to add a dependent.
Pre-Conditions	Patient must be logged in.

Post-Conditions	Family members are added successfully.
Normal Flow	1. Patient logs in → 2. Navigates to profile → 3. Adds family details → 4. Confirms addition
Alternative Flow	Invalid details provided
Exceptions	Server error, no internet, or duplicate entries
Business Rules	Each patient can add a limited number of dependents

3.5.11 Approve Doctor Registration

The “Approve Doctor Registration” use case allows the admin to verify and approve doctor accounts within the Health Cloud system. As presented in Table 3.11, this use case is triggered whenever a doctor submits a registration request. The pre-condition requires that the doctor has submitted valid documents. Normally, the admin logs in, reviews the request, and approves or rejects it, activating the doctor’s account. Alternative flows account for cases where approval is delayed for further verification, while exceptions include server issues or invalid documents. Business rules ensure that only verified doctors are approved, maintaining system integrity and quality of care.

Table 3.11: Approve Doctor Registration Use Case

Use Case Attribute	Details
Use Case Name	Approve Doctor Registration
Primary Actors	Admin
Description	Admin verifies and approves doctor accounts.
Trigger	A doctor submits a registration request.

Pre-Conditions	Doctor has submitted valid documents.
Post-Conditions	Doctor's account is activated.
Normal Flow	1. Admin logs in → 2. Reviews request → 3. Approves or rejects
Alternative Flow	Approval is delayed for further verification
Exceptions	Server issues, invalid documents
Business Rules	Only verified doctors are approved

3.5.12 Manage Finance

The “Manage Finance” use case allows the admin to track payments, revenue, and invoices within the Health Cloud system. As outlined in Table 3.12, this use case is triggered whenever a new transaction occurs. The pre-condition requires that the system has financial records. Normally, the admin logs in, navigates to the finance dashboard, and views transactions to ensure all financial details are updated. Alternative flows occur when transaction records are missing, while exceptions include payment gateway errors. Business rules enforce that all transactions must be logged accurately for proper auditing and financial management.

Table 3.12: Manage Finance Use Case

Use Case Attribute	Details
Use Case Name	Manage Finance
Primary Actors	Admin
Description	The admin can track payments, revenue, and invoices.
Trigger	A new transaction occurs.
Pre-Conditions	System has financial records.

Post-Conditions	Finance details are updated.
Normal Flow	1. Login → 2. Navigate to finance dashboard → 3. View transactions
Alternative Flow	Transaction records are missing
Exceptions	Payment gateway error
Business Rules	All transactions must be logged

3.5.13 Manage Users

The “Manage Users” use case enables the admin to add, suspend, or remove users within the Health Cloud system. As shown in Table 3.13, this use case is triggered when user activity is monitored. The pre-condition requires that the user exists in the system. Normally, the admin logs in, navigates to user management, and edits the user’s status. Alternative flows handle cases where an unauthorized admin tries to modify data, while exceptions include server errors or lack of internet connectivity. Business rules enforce that only authorized admins with proper permissions can modify user accounts, ensuring system security.

Table 3.13: Manage Users Use Case

Use Case Attribute	Details
Use Case Name	Manage Users
Primary Actors	Admin
Description	Admin can add, suspend, or remove users.
Trigger	User activity is monitored.
Pre-Conditions	User exists in the system.
Post-Conditions	User status is updated.

Normal Flow	1. Login → 2. Navigate to user management → 3. Edit user status
Alternative Flow	Unauthorized admin tries to modify data
Exceptions	Server error, no internet
Business Rules	Admins must have permission to modify users

3.5.14 Monitor System Logs

The “Monitor System Logs” use case allows admins to track system activity and review security logs within the Health Cloud system. As indicated in Table 3.14, this use case is triggered whenever an issue is reported. The pre-condition requires that system logs are available. Normally, the admin logs in, navigates to system logs, and reviews the reports. Alternative flows account for situations where no logs are available, while exceptions include corrupt logs. Business rules ensure that all logs are stored securely to maintain system accountability and security.

Table 3.14: Monitor System Logs Use Case

Use Case Attribute	Details
Use Case Name	Monitor System Logs
Primary Actors	Admin
Description	Admins can track system activity and security logs.
Trigger	An issue is reported.
Pre-Conditions	Logs are available.
Post-Conditions	System logs are reviewed.
Normal Flow	1. Login → 2. Navigate to system logs → 3. Review reports

Alternative Flow	No logs available
Exceptions	Corrupt logs
Business Rules	Logs must be stored securely

3.5.15 Manage Appointments

The “Manage Appointments” use case allows admins and staff to manage and reschedule appointments within the Health Cloud system. As shown in Table 3.15, this use case is triggered whenever a patient or doctor requests a change in an existing appointment. The pre-condition requires that appointment records already exist in the system. Normally, the user logs in, navigates to the appointment management section, and modifies the schedule as needed. Alternative flows handle situations where appointment changes are restricted, while exceptions include server issues or invalid modifications. Business rules ensure that only authorized personnel are permitted to modify appointment schedules, maintaining accurate and secure scheduling.

Table 3.15: Manage Appointments Use Case

Use Case Attribute	Details
Use Case Name	Manage Appointments
Primary Actors	Admin, Staff
Description	Admins and staff can manage and reschedule appointments.
Trigger	A patient or doctor requests a change.
Pre-Conditions	Appointment records exist.
Post-Conditions	Appointments are updated.
Normal Flow	1. Login → 2. Navigate appointment management → 3. Modify schedule

Alternative Flow	Appointment changes are restricted
Exceptions	Server issues, invalid modifications
Business Rules	Only authorized personnel can modify schedules

3.6 Functional Requirements

Functional requirements define the specific behaviors and operations that the Health Cloud system must provide to satisfy the needs of its users. These requirements outline the essential capabilities that the system must possess to manage patient health data, appointments, communication between patients, doctors, and staff, as well as administrative tasks effectively. The Health Cloud system is designed to offer a secure, reliable, and user-friendly environment that ensures data integrity, privacy, and efficient healthcare service management.

3.6.1 User Management

The Health Cloud system allows patients, doctors, and administrators to register using valid credentials such as email, phone number, and secure passwords. Registration is guided to ensure all required fields are completed correctly, and login is secured with role-based authentication. Patients access personal records and appointments, doctors access patient data and consultation tools, and administrators manage system functions. Security measures like hashed passwords and optional OTP verification protect sensitive information, and repeated failed logins may trigger temporary account locks.

Users can manage and update their profiles after registration. Patients can maintain contact details, health preferences, and add family members for shared tracking. Doctors provide professional details such as specialization, experience, and availability, which can be updated dynamically. The system validates all changes and sends automated notifications to keep users informed of profile updates or verification needs.

Account verification ensures authenticity. Administrators review doctor registrations, validate documents, and approve or reject accounts, while the system notifies users of their registration

status. Patients may verify accounts via email or phone. All verification activities are logged to maintain transparency and accountability.

3.6.2 Patient Records Management

The system provides secure access to patient health records, enabling patients to view their complete medical history, including consultations, prescriptions, test results, and treatment plans. A clear, user-friendly interface helps patients monitor their health and make informed decisions.

Doctors can access patient records during consultations to provide accurate diagnoses and treatment. Strict access control ensures only authorized users can view or modify records, and all access and modifications are logged for accountability and compliance.

Data security is critical; all records are encrypted and protected against unauthorized access. Any unauthorized attempts are detected, logged, and flagged to administrators to maintain confidentiality and integrity.

3.6.3 Appointments Management

The system allows patients, doctors, staff, and administrators to schedule and manage appointments efficiently. Patients or staff can book appointments using an interface that displays available doctor slots, preventing overlaps and ensuring optimal resource use.

Doctors can accept or decline requests, with immediate notifications sent to patients. Administrators and staff can modify, reschedule, or cancel appointments while enforcing access control to prevent unauthorized changes.

Automated reminders and alerts keep users informed of upcoming appointments or changes, improving coordination, adherence to schedules, and overall service efficiency.

3.6.4 Prescriptions and Medications

Doctors can issue prescriptions securely after consultations, and prescriptions are linked to patient records. Each prescription contains detailed information about medication, dosage, and duration. Patients can access and review prescriptions to follow their treatment accurately.

The system ensures that only licensed doctors can issue prescriptions, preventing misuse. Patients can request clarifications directly through the system to enhance understanding and engagement in their care.

Prescription data is securely stored, encrypted, and logged for compliance and safety. These measures protect sensitive medical information and ensure the system provides reliable, legally compliant prescription management.

3.6.5 Payment and Financial Management

The Health Cloud system allows patients to pay for consultations, prescriptions, and other services through a secure online platform. Multiple payment methods, including credit cards, bank transfers, and digital wallets, are supported, and the interface guides users through the process while providing confirmation of successful transactions.

All financial transactions are encrypted and securely logged to protect sensitive information. The system validates payments in real-time, issues receipts or notifications, and alerts patients and administrators of any failed or incomplete transactions to resolve issues promptly.

Administrators have access to robust finance management tools, enabling them to track revenue, review invoices, and generate comprehensive reports. Detailed transaction records facilitate auditing, monitoring of revenue streams, detection of discrepancies, and compliance with accounting standards, ensuring a secure and transparent financial system.

3.6.6 Notifications and Alerts

The system provides a comprehensive notification mechanism to keep users informed of important updates. Patients and doctors receive reminders for upcoming appointments via email, SMS, or in-app notifications, reducing missed consultations and ensuring timely healthcare delivery.

Any changes to appointments automatically trigger alerts for both parties to maintain transparency and prevent scheduling conflicts. Notifications also cover payment confirmations, profile updates, prescription approvals, and account modifications. Optional critical health alerts can notify patients about abnormal test results or vital health information, enabling timely interventions.

3.6.7 Admin and System Management

Administrators have comprehensive tools to manage users, monitor system activity, and enforce security policies. They can add, suspend, or remove users, ensuring that only authorized individuals access the system. Monitoring user activity helps detect suspicious behavior and maintain compliance.

All actions and critical events are logged for auditing and accountability. Administrators can review logs to track usage, identify anomalies, and take corrective actions. Role-based access control ensures that users can only access features and data relevant to their role, protecting sensitive information while allowing efficient task execution.

3.6.8 Data Reporting and Analytics

The Health Cloud system provides reporting and analytics to support informed decision-making. Doctors can access health reports summarizing patient histories, consultations, prescriptions, and appointments, enabling them to track diagnosis trends, treatment outcomes, and plan follow-ups effectively [2].

Administrators can generate financial reports to track payments, revenue streams, and resource allocation. Analytics on system usage, appointment trends, and user engagement help optimize performance and improve user experience.

All reporting and analytics features rely on secure data handling and real-time processing to ensure accurate, up-to-date insights. Automated report generation and alerts further support proactive management of healthcare services and administrative operations.

3.7 Non-Functional Requirements

Non-functional requirements define the quality attributes, performance standards, and operational constraints that the Health Cloud system must meet. While functional requirements describe what the system does, non-functional requirements specify how the system performs and the standards it must uphold in terms of reliability, security, usability, and efficiency. These requirements are critical for ensuring that the Health Cloud system operates effectively under real-world conditions, meets user expectations, and complies with healthcare regulations and standards.

3.7.1 Performance

The Health Cloud system must be capable of handling multiple simultaneous users without noticeable delays, ensuring fast response times for all operations, including accessing medical records, booking appointments, and processing payments. The system should be optimized for high performance so that queries on patient data, prescription history, and financial transactions are executed efficiently. Adequate server resources, database optimization, and load balancing must be implemented to maintain consistent performance even during peak usage hours.

3.7.2 Scalability

The system must be scalable to accommodate growth in the number of users, patient records, appointments, and medical data over time. As more patients, doctors, and administrators use the platform, the system should be able to scale horizontally or vertically without affecting performance. This ensures that Health Cloud can expand its services to larger healthcare networks or multiple clinics while maintaining responsiveness and stability.

3.7.3 Security

Security is a critical non-functional requirement for Health Cloud due to the sensitivity of patient health information and financial transactions. The system must implement encryption for data at rest and in transit, ensuring that all patient records, prescriptions, and payment details are protected against unauthorized access. Role-based access control must be strictly enforced to ensure that only authorized users can view or modify sensitive information. Additionally, the system must support secure authentication methods, such as strong passwords, OTP verification, and multi-factor authentication. Regular security audits and monitoring must be conducted to detect and mitigate potential vulnerabilities.

3.7.4 Reliability and Availability

Health Cloud must provide high reliability and ensure continuous availability for all users, including patients, doctors, staff, and administrators. The system should have minimal downtime, and robust error-handling mechanisms must be in place to manage server failures, network disruptions, or unexpected exceptions. Redundancy, fault tolerance, and real-time monitoring of

system health are essential to maintain 24/7 availability. Automatic failover procedures and backup mechanisms must be implemented to ensure uninterrupted access and protection of critical data.

3.7.5 Usability

The system must offer an intuitive and user-friendly interface suitable for users with varying levels of technical expertise. Patients, doctors, and administrators should be able to navigate the system easily to perform tasks such as viewing medical records, booking appointments, issuing prescriptions, or managing users. Clear navigation, consistent design, accessible forms, and contextual help features are required to enhance the user experience and reduce errors.

3.7.6 Maintainability

Health Cloud must be designed for ease of maintenance and updates. The system should support modular architecture, allowing components to be updated or replaced without affecting other functionalities. Clear documentation, coding standards, and version control practices must be followed to facilitate efficient maintenance, debugging, and future enhancements.

3.7.7 Compliance

The system must comply with healthcare regulations, data protection laws, and industry standards such as HIPAA (or local equivalents) to ensure that patient information is handled legally and ethically. All operations, from storing medical records to processing payments, must adhere to these compliance standards to protect patients and healthcare providers from legal and operational risks.

3.7.8 Backup and Recovery

The Health Cloud system must implement regular data backups and provide reliable recovery mechanisms. Backups should be automated and stored securely, ensuring that patient records, prescriptions, financial data, and appointment details can be restored in the event of system failure, accidental deletion, or data corruption. Recovery procedures must be tested periodically to verify effectiveness and minimize downtime during emergencies.

3.7.9 Performance Monitoring and Logging

The system must include mechanisms for monitoring performance, logging user activity, and detecting anomalies. Real-time monitoring of server load, database performance, and application errors ensures that potential issues are identified and addressed proactively.

Chapter 4

DESIGN AND ARCHITECTURE

4 Design And Architecture

The Design and Architecture chapter of the Health Cloud system explains how the software is structured, how its components interact, and how data flows throughout the system. This chapter provides a detailed overview of the system's architecture, data representation, process flows, and design models. It highlights the technical decisions made to ensure performance, security, scalability, and maintainability. By presenting the system design clearly, this chapter demonstrates how Health Cloud meets both functional and non-functional requirements while delivering a reliable healthcare management platform.

4.1 System Architecture

The system architecture defines the overall structure of Health Cloud and illustrates how its main components work together to deliver healthcare services efficiently [6]. It ensures that each module performs its specific tasks while interacting seamlessly with others to maintain smooth operations and data integrity.

4.1.1 Architecture Style

Health Cloud is developed using a layered client-server architecture following the MVC (Model-View-Controller) design pattern. The client layer consists of web and mobile interfaces that allow patients, doctors, staff, and administrators to interact with the system. The server layer manages business logic, authentication, data processing, and communication with the database. This separation of concerns enhances maintainability, supports scalability as the number of users grows, and ensures secure, organized handling of system operations.

4.1.2 Components

The Health Cloud system is composed of several interconnected components, each designed to perform a specific function within the overall architecture. These components work collaboratively to ensure smooth system operation, secure data handling, and efficient healthcare service delivery. By dividing the system into modular parts, Health Cloud maintains clarity in functionality,

supports easier maintenance, and allows individual components to evolve without affecting the entire system.

4.1.2.1 User Interface (UI)

The UI provides web and mobile access to all user types patients, doctors, staff, and administrators. It is designed to be intuitive and responsive, ensuring users can navigate features such as registration, profile management, appointment scheduling, prescription viewing, and financial transactions with ease. The UI serves as the primary point of interaction between the user and the system.

4.1.2.2 Authentication Service

The authentication service manages user registration, login, and role-based access control. It ensures that only authorized users can access the system according to their roles, implementing security measures such as hashed password storage, OTP verification, and monitoring of failed login attempts. This component safeguards sensitive data and prevents unauthorized access.

4.1.2.3 Patient Records Module

This module stores and manages comprehensive patient health information, including medical histories, prescriptions, diagnostic results, and treatment plans. It implements strict access control to ensure that only patients and their authorized doctors can view or modify records. Audit logs track all access and changes, maintaining transparency and compliance with healthcare regulations.

4.1.2.4 Appointments Module

The appointments module facilitates booking, rescheduling, and managing appointments. It displays available time slots based on doctor schedules and ensures no conflicts occur. Doctors can accept or decline requests, while notifications keep patients informed. Administrators and staff can modify schedules when necessary, ensuring efficient allocation of resources and smooth workflow management.

4.1.2.5 Payment Module

The payment module allows patients to make secure online payments for consultations, prescriptions, and other services. It supports multiple payment methods such as credit cards, bank

transfers, and digital wallets. The module validates transactions, generates receipts, and logs all financial activities for auditing and security purposes.

4.1.2.6 Admin Dashboard

The admin dashboard provides tools for administrators to manage users, monitor system activity, and generate reports. It allows oversight of appointments, financial transactions, and system logs, enabling effective management, security enforcement, and operational decision-making.

4.1.3 Interactions

All components communicate primarily through RESTful APIs, ensuring efficient and standardized data exchange. The UI sends requests to the server layer, which processes data, interacts with the database, and responds with relevant information. For instance, when a patient books an appointment, the UI sends the request to the appointment's module, which verifies doctor availability, updates records, and sends notifications to both the doctor and patient. All communication is secured using HTTPS protocols and authentication tokens to protect data integrity and user privacy.

4.1.4 System Architecture Diagram

The system architecture diagram (Figure 4.1) illustrates the overall structure and interaction of the Hospital Management System components. It shows how users including patients, doctors, staff, and administrators interact with the application through web and mobile interfaces. The diagram highlights the flow of information between the user interface, the application server, and the database.

The architecture is designed to ensure smooth communication between modules and efficient data management. When a user performs an action, such as logging in, registering, or booking an appointment, the request is sent to the application server. The server processes the request, interacts with the database to retrieve or store data, and then returns the response to the user interface. This structure provides a clear separation of concerns, improves system scalability, and enhances maintainability.

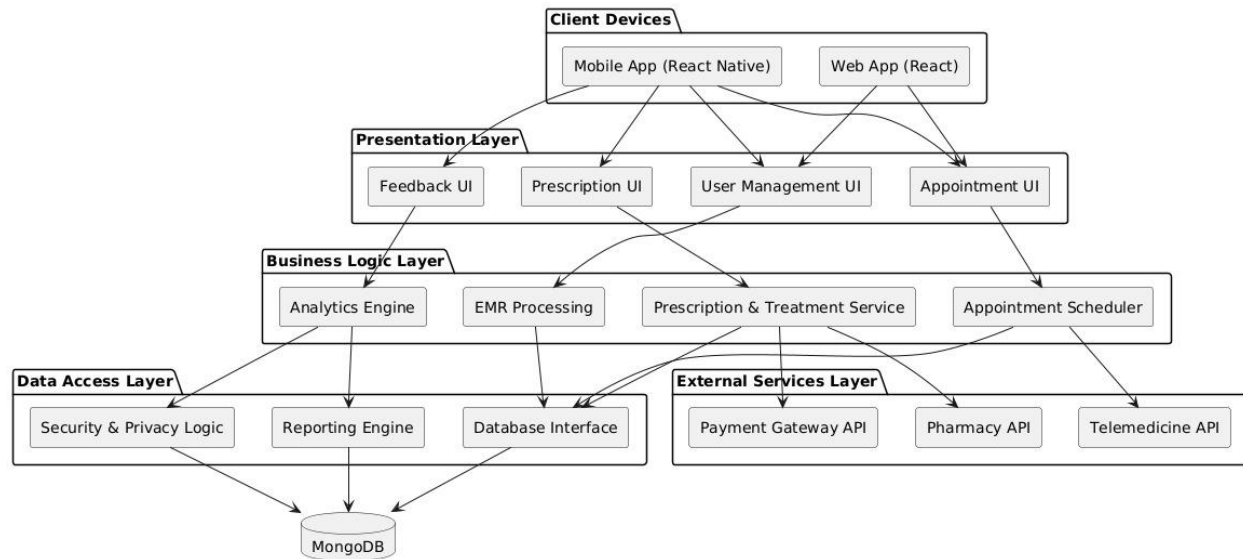


Figure 4.1: System Architecture Diagram

4.2 Data Representation

The Data Representation section explains how information is structured, stored, and managed within the Health Cloud system. It covers the database design approach, entity relationships, data flow between modules, and the organization and security of stored data. This ensures that all system components can efficiently access and process the information they need while maintaining accuracy, consistency, and privacy.

4.2.1 Database Design

Health Cloud uses a NoSQL database (MongoDB) due to its flexibility, scalability, and ability to handle large volumes of semi-structured medical data. Since health records vary greatly in format such as prescriptions, test results, medical histories, and appointment logs MongoDB allows storing this information in dynamic document structures. This design supports rapid read/write operations, horizontal scalability, and easy integration with cloud-based healthcare solutions. The decision to use MongoDB also helps maintain system responsiveness even when managing thousands of patients and real-time updates from doctors, appointment modules, and payment services.

4.2.2 Entity–Relationship Representation

Although MongoDB does not use traditional relational tables, the system follows a structured entity model to maintain consistency. Each entity (such as Patient, Doctor, Appointment, Payment, and Prescription) is stored as a document with well-defined attributes. Relationships are implemented using document references or embedded documents depending on access frequency and data size. An ER-style diagram (Figure 4.4) is used to illustrate how major entities connect for example, each patient has multiple appointments, each appointment links to a doctor, and prescriptions are associated with both patients and doctors. This structure ensures clarity in how data is organized and accessed across modules.

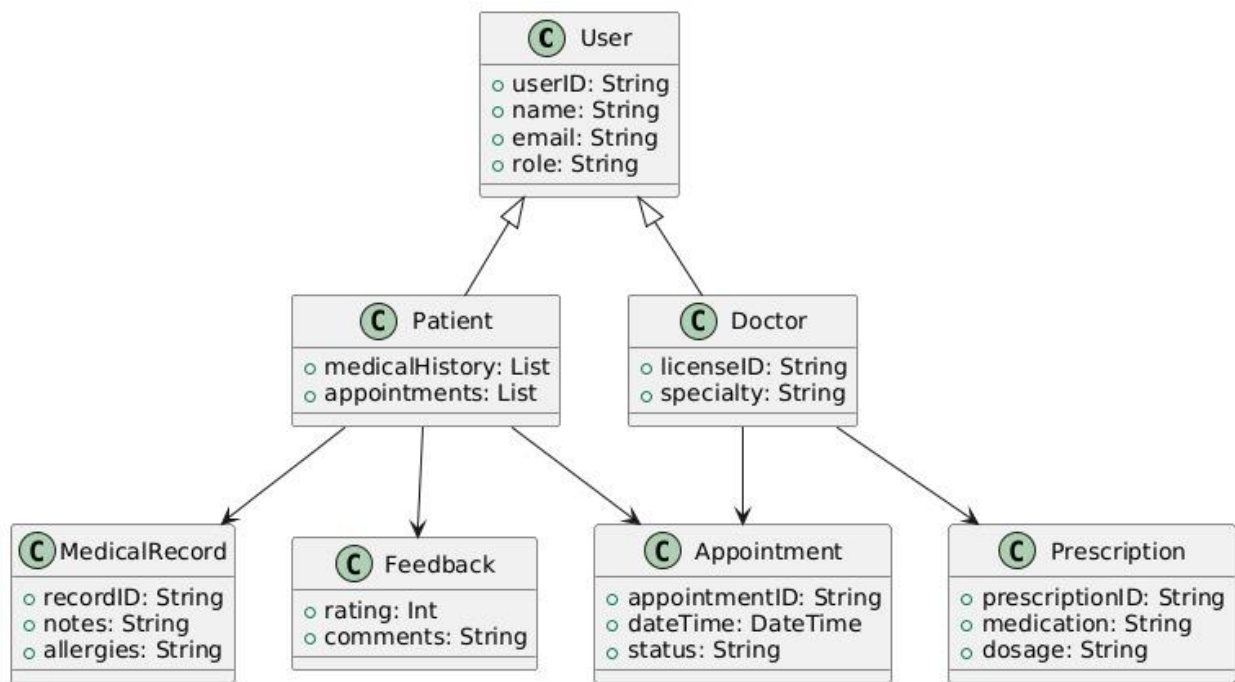


Figure 4.2: Entity–Relationship Diagram of the System

The second representation (Figure 4.5) provides a more detailed breakdown of embedded documents and reference-based links, showing how MongoDB manages nested structures for high-frequency data access such as prescriptions and appointment records.

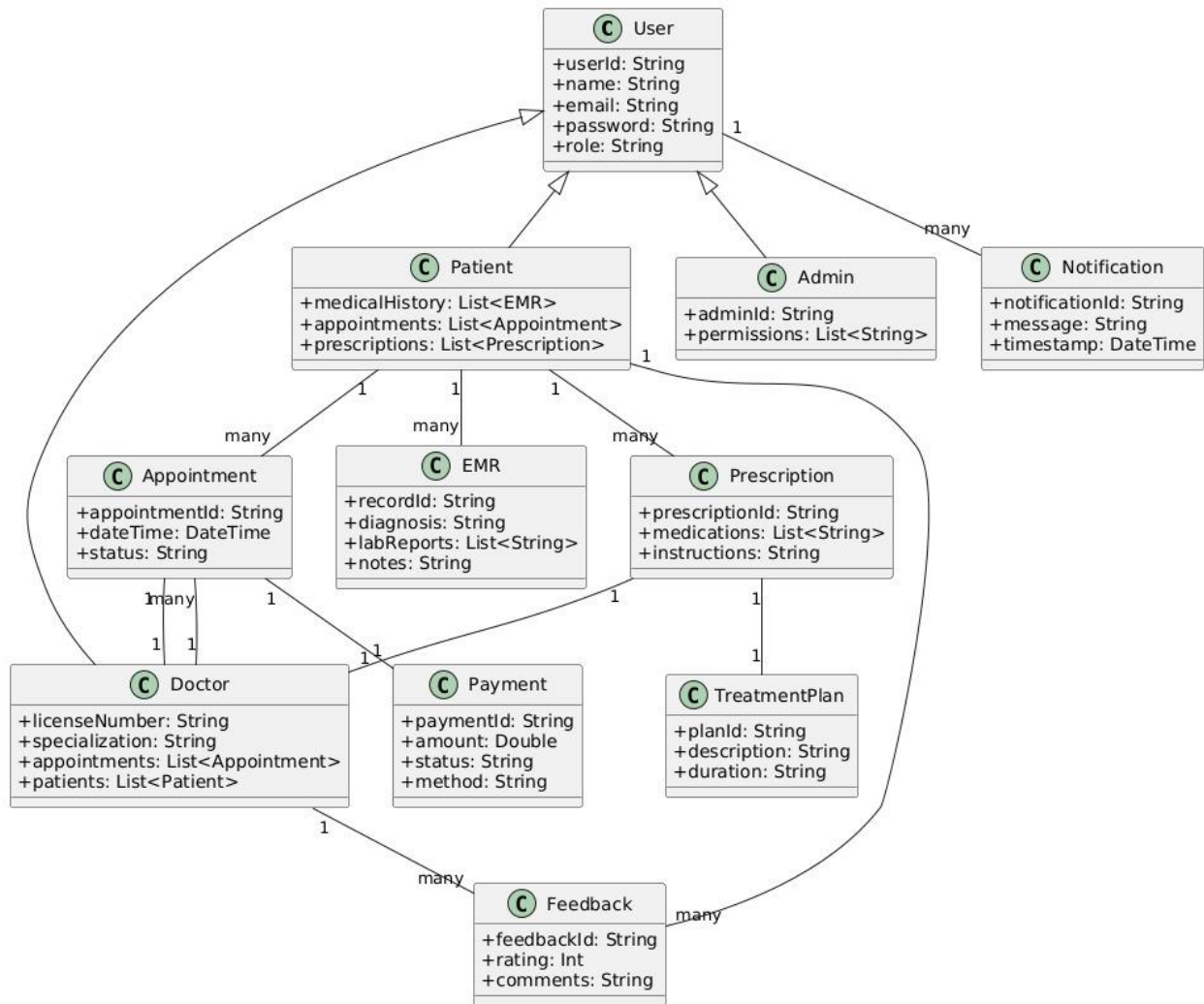


Figure 4.3: Detailed Document Relationship Structure in MongoDB

4.2.3 Data Flow

Data in the Health Cloud system moves through a well-defined pathway starting from the user interface and ending in secure storage. When a patient books an appointment, the information is captured in the presentation layer, validated in the business logic layer, and then saved to the database. Doctors later retrieve this stored data through their dashboards for consultations. Similarly, payment-related information flows from the user interface to the payment module and then into secure financial records. These processes collectively ensure smooth coordination between appointments, prescriptions, user accounts, and analytics, as illustrated in Figure 4.2.

4.2.3.1 Sequence Diagram (Data Flow Across Layers)

The sequence diagram shown in Figure 4.2 explains how data moves between the presentation layer, business logic layer, database layer, and external services during major operations such as appointment booking, EMR viewing, prescription processing, and payment handling.

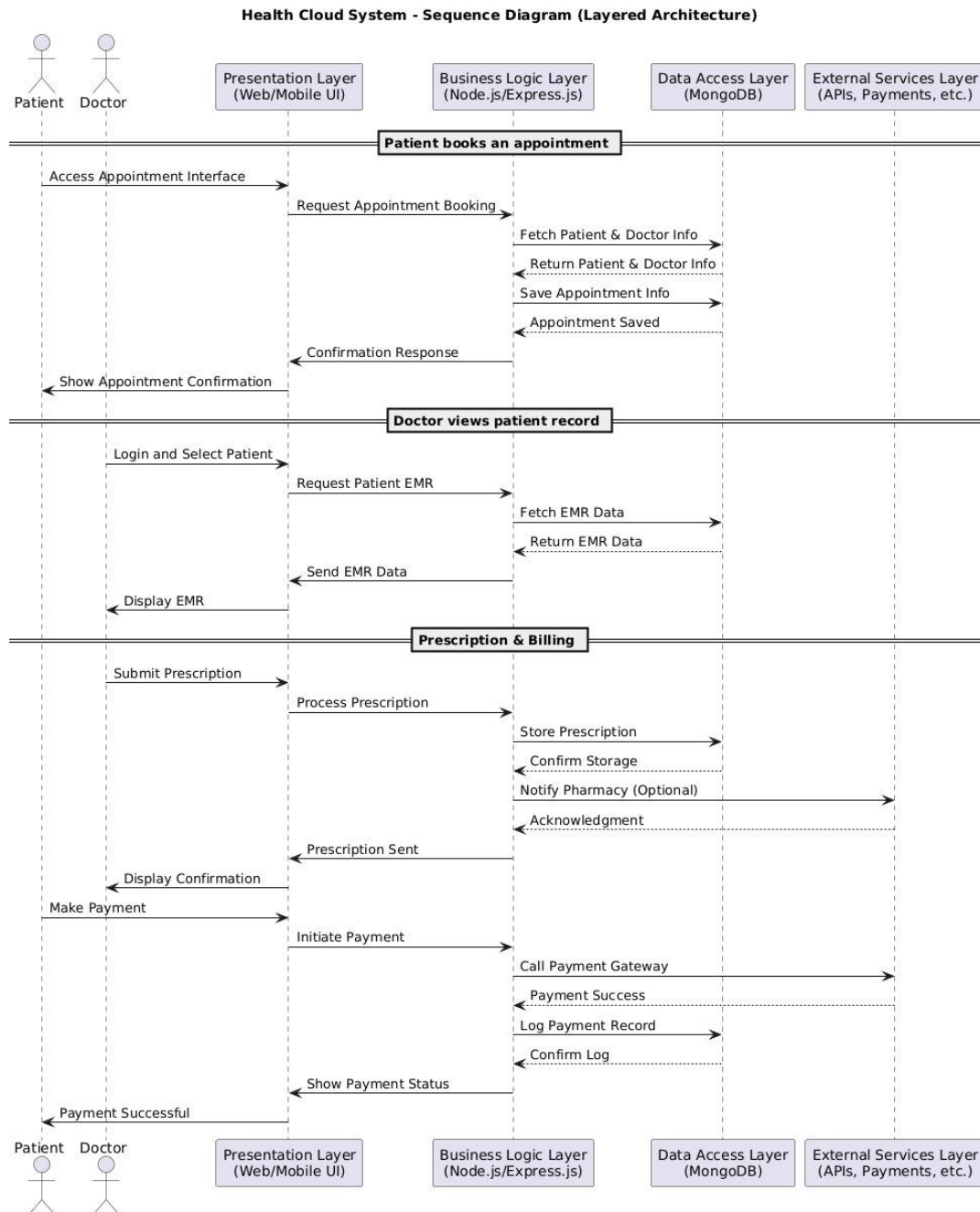


Figure 4.4: Layered sequence diagram

4.2.3.2 Activity Diagram (Overall System Workflow)

The activity diagram in Figure 4.3 summarizes the entire workflow from patient login to feedback submission. It illustrates how each action such as booking appointments, issuing prescriptions, and making payments results in data being created, processed, and stored across various modules of the system.

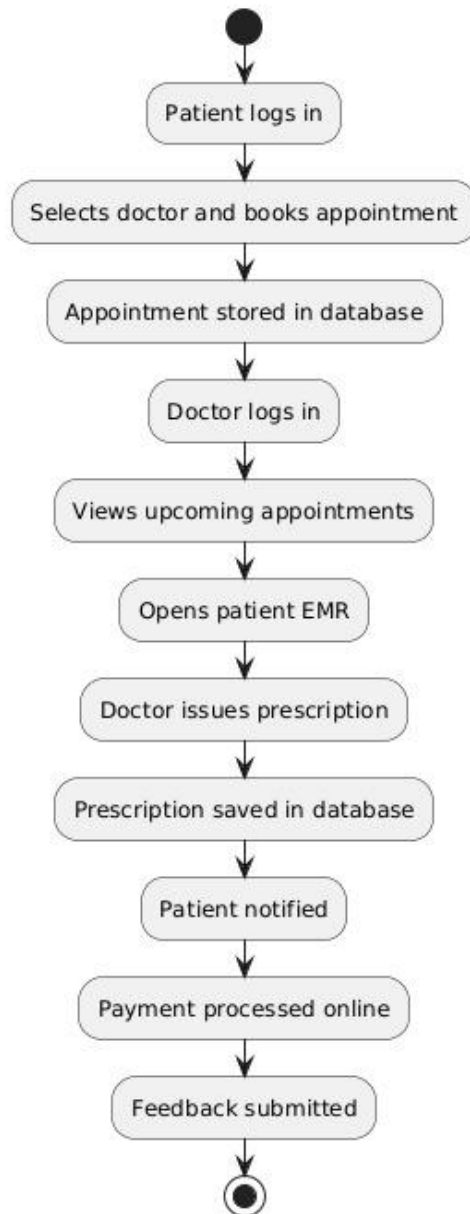


Figure 4.5: Vertical patient-journey activity diagram

4.2.4 Entity Descriptions and Constraints

Each entity within the database serves a specific purpose. Patient entities store personal details, medical history, and account information. Doctor entities maintain professional profiles, specializations, availability schedules, and consulted patients. Appointment entities include patient IDs, doctor IDs, time slots, status, and notes. Prescription entities record medication details and are linked to patient and doctor IDs. Payment entities include transaction IDs, amounts, timestamps, and verification details. Each entity includes constraints such as unique identifiers, required fields, validation rules, and role-based restrictions. Data integrity is maintained through controlled data updates, validation at the business logic layer, and strict access permissions. Security is enforced through encryption, audit logs, and database access restrictions to protect sensitive health information.

4.3 Process Flow / Representation

The Process Flow section explains how key operations are carried out in the Health Cloud system. Even when diagrams are not included, each workflow is described clearly through structured explanations. These flows help readers understand the triggers, steps, and outcomes involved in major system processes.

4.3.1 Login and Authentication Process

The login and authentication process begins when a user enters their credentials through the Health Cloud interface. The system receives the input and verifies it by comparing the credentials with the records stored in the authentication database. If the information matches, the system checks the user's role to determine the correct dashboard to load. After successful verification, a secure session token is generated, allowing the user to access the system without repeatedly logging in. If the credentials are incorrect, the system rejects the login attempt and prompts the user to try again, ensuring secure access control.

4.3.2 Appointment Booking Process

The appointment booking process starts when a patient selects a doctor and initiates the scheduling option. The system retrieves the doctor's availability from the database and displays the open time

slots to the patient. Once a slot is chosen and confirmed, the system records the appointment, updates the doctor's schedule, and stores the information in the database. Notifications are automatically sent to both the patient and doctor, informing them of the confirmed appointment. If another user has already booked the selected time, the system alerts the patient and requests them to choose a different slot, ensuring smooth and conflict-free scheduling.

4.3.3 Prescription Generation Process

The prescription generation process begins when a doctor accesses a patient's medical profile and enters diagnosis details and recommended treatments. The doctor fills in the required fields, adds medication information, and submits the prescription. The system then stores the prescription securely in the patient's health record, making it immediately accessible through the patient dashboard. A notification is generated to inform the patient, and the prescription becomes available for future reference, pharmacy integration, or follow-up consultations.

4.3.4 Payment Processing Workflow

The payment processing workflow is triggered when a patient proceeds to make a payment for a consultation or service. The system sends the transaction information to the external payment gateway for verification and processing. After the payment is successfully completed, a receipt is generated and stored in the financial records. The system updates the transaction logs and grants the patient access to the paid services or documents. If the payment fails or is declined, a notification is displayed to the user, prompting them to retry or choose an alternate payment method.

4.4 Design Models

Design models describe the structural and behavioral aspects of the Health Cloud system through conceptual UML-like representations. Even without images, each model is thoroughly explained to reflect system behavior, module interaction, and architectural decisions.

4.4.1 Class Model Description

The class model defines the main structural elements of the Health Cloud system by outlining the central classes and their relationships. Classes such as Patient, Doctor, Appointment, Prescription, and Payment contain relevant attributes and operations that describe their behavior within the system. The Patient class maintains personal and medical information, while the Doctor class stores professional details. The Appointment class links patients and doctors through scheduling information, and the Prescription class connects diagnosis data to both parties. These relationships demonstrate how objects interact and how data is exchanged across different modules.

4.4.2 Sequence Model Description

The sequence model illustrates the communication between system components during important operations. For example, during the appointment scheduling process, the sequence begins with the patient sending a request for available time slots. The appointment scheduler retrieves availability information from the database and returns it to the user interface. After the patient confirms a selection, the scheduler stores the appointment details and notifies all involved parties. This sequence highlights the order of interactions and clarifies how messages flow between components during each step of the process.

4.4.3 Activity Model Description

The activity model explains the workflows within the system by describing the actions, decisions, and outcomes that occur during various processes. In the login activity, the user enters their credentials, and the system checks the user type before validating the details. If the credentials are accurate, the appropriate dashboard is loaded; otherwise, an error message is shown. This representation demonstrates how the system handles different scenarios and ensures that all necessary steps are followed according to predefined rules and logic.

4.4.4 Component Model Description

The component model describes how the major modules of the Health Cloud system are organized and interconnected. Components such as the user interface, authentication service, appointment scheduler, patient record manager, and payment processor operate independently while communicating through secure APIs. External services, including payment gateways.

Chapter 5

IMPLEMENTATION

5 Implementation

This chapter explains how the Health Cloud system was implemented at the technical level. It does not include source code but provides descriptions of the algorithms, external APIs, and user interface decisions that were used to build the system. The goal of this chapter is to show how the system's design was transformed into a functional software solution while maintaining efficiency, accuracy, and security.

5.1 Algorithms

This section describes the main algorithms used in the Health Cloud system. Each algorithm represents the internal logic behind system features and explains how different functions operate. These descriptions may be presented in pseudocode or natural language to clearly communicate the workflow behind major modules.

5.1.1 Authentication Algorithm

The authentication algorithm ensures that only legitimate users can access the Health Cloud system. It protects sensitive patient and administrative data by verifying credentials and granting role-based access. This algorithm is crucial for maintaining system security and preventing unauthorized access.

Steps:

1. User enters email and password.
2. Retrieve stored credentials for the given email from the database.
3. Compare the entered password with the stored password.
4. If credentials match:
 - a. Generate a secure authentication token.
 - b. Identify user role (Patient / Doctor / Admin).
 - c. Redirect user to the corresponding dashboard.
5. If credentials do not match:
 - a. Record the failed login attempt.

- b. If failed attempts exceed allowed limit, temporarily lock the account.
- c. Notify the user of invalid login.

5.1.2 Appointment Scheduling Algorithm

This algorithm allows patients to book appointments efficiently while preventing scheduling conflicts. It ensures that doctors' time slots are managed accurately and that notifications are sent to both parties, improving overall workflow and patient experience.

Steps:

1. Patient selects desired date and time for an appointment.
2. Retrieve doctor's schedule for the selected date.
3. Check if the selected time slot is available and has no conflicts.
4. If slot is available:
 - a. Confirm appointment.
 - b. Save appointment details in the database.
 - c. Notify both patient and doctor.
5. If slot is unavailable:
 - a. Prompt patient to select another available slot.

5.1.3 Prescription Generation Algorithm

The prescription generation algorithm allows doctors to create accurate medical prescriptions within the system. It ensures that prescriptions are only issued by authorized personnel and that they are properly linked to the patient's record and relevant appointments for future reference.

Steps:

1. Doctor opens patient record and selects "Create Prescription."
2. Enter medication details, dosage, and treatment duration.
3. Verify that all required fields are completed.
4. Check doctor authorization to issue prescriptions.

5. If authorized and all data is valid:
 - a. Save prescription in patient's medical record.
 - b. Link prescription to the appointment.
6. Provide confirmation to doctor and make prescription accessible for future reference.

5.1.4 Payment Processing Algorithm

This algorithm manages the secure processing of online payments for consultations and other services. It ensures that all transactions are validated by a trusted payment gateway and that successful payments are recorded while failures are logged for review.

Steps:

1. Patient initiates payment and enters payment details.
2. Send payment information to the external payment gateway.
3. Gateway validates transaction and returns a response.
4. If payment is successful:
 - a. Store payment details in the database.
 - b. Generate and provide receipt to patient.
5. If payment fails:
 - a. Notify patient of transaction failure.
 - b. Log failed transaction for administrative review.

5.2 External APIs

The Health Cloud system leverages several external APIs to provide advanced functionality beyond the core MERN stack capabilities. These APIs enhance features such as secure payment processing, email and WhatsApp notifications, location services, and user authentication, improving overall system efficiency and user experience. In the MERN architecture, APIs are primarily consumed by the backend (Node.js/Express) to handle requests, validate data, and interact with the database (MongoDB), while the frontend (React) dynamically renders responses and provides interactive features to users.

The table below summarizes the key APIs used in the project, their purpose, and where they are integrated within the system. These include payment gateway APIs like Stripe or PayPal for handling transactions, email and WhatsApp APIs for sending notifications and updates to patients and doctors, Google Maps API for location services, and JWT for secure authentication and session management. Each API is carefully integrated to ensure security, reliability, and seamless interaction between the frontend and backend components.

Table 5.1: Details of APIs Used in the Project

Name of API	Description of API	Purpose of Usage	Function/Class Name in which Used
Payment Gateway API	Facilitates secure online payments (e.g., Stripe, PayPal)	Process patient payments for consultations or services	<code>PaymentProcessor</code> (Backend Node.js/Express)
Email Notification API	Sends email notifications to users	Notify patients and doctors about appointments, prescriptions, and system alerts	<code>NotificationService</code> (Backend Node.js/Express)
WhatsApp Notification API	Sends WhatsApp messages to users	Notify patients and doctors of appointments, reminders, or alerts	<code>NotificationService</code> (Backend Node.js/Express)
Google Maps API	Provides location and map services	Display clinic locations, find nearby doctors,	<code>LocationService</code> (Frontend React for UI,

		and calculate distances	Backend for API requests)
JWT Authentication API	Generates and validates tokens for secure sessions	Securely authenticate users and maintain session state across frontend and backend	AuthController (Backend Node.js/Express, used in login/signup flows)

5.3 User Interface

The user interface (UI) of the Health Cloud system is designed to provide a seamless, intuitive, and user-friendly experience for patients, doctors, staff, and administrators. It focuses on clarity, accessibility, and efficiency, ensuring that users can navigate the system with minimal learning effort. Consistent layout patterns, well-structured menus, and clearly labeled icons help users perform tasks such as booking appointments, managing patient records, generating prescriptions, and monitoring system activity. The design follows modern UI principles with responsive layouts, allowing smooth usage across devices including desktops, tablets, and mobile phones. Visual hierarchies, appropriate color schemes, and simple forms enhance usability, while error messages, confirmation dialogs, and interactive feedback ensure better user control and guidance. Overall, the UI is developed to support fast task completion, reduce user errors, and improve the overall experience of digital healthcare management.

The Home Page shown in Figure 5.1 of the Patient Application, which acts as the main interface through which users access all system features. The layout is designed to be clean and user-friendly, ensuring that patients can quickly navigate to key options such as booking appointments, viewing prescriptions, checking notifications, and managing their profile. The prominent icons and structured sections help users interact with the application efficiently, making the overall experience smooth and intuitive.

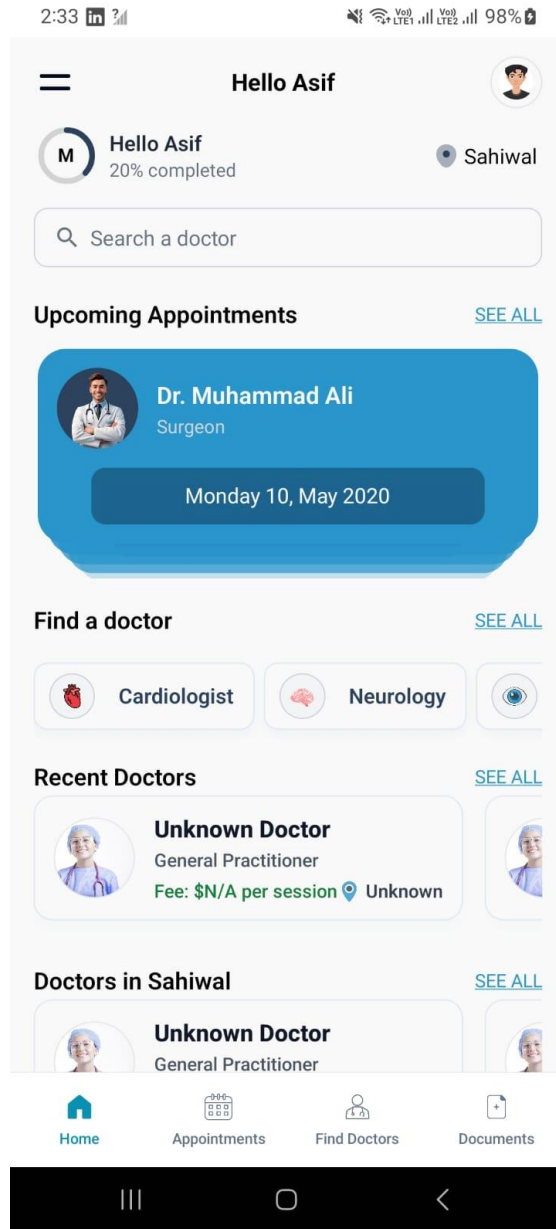


Figure 5.1: Home page of doctor app

Figure 5.2 displays the Doctor Profile Review Form within the Patient Application, where patients can provide feedback after a consultation. This form allows users to rate the doctor, write comments, and share their experience in a structured manner. The interface is simple and easy to understand, enabling patients to submit reviews quickly. This feature plays an important role in maintaining service quality by collecting valuable patient insights that assist administrators and doctors in improving healthcare delivery.

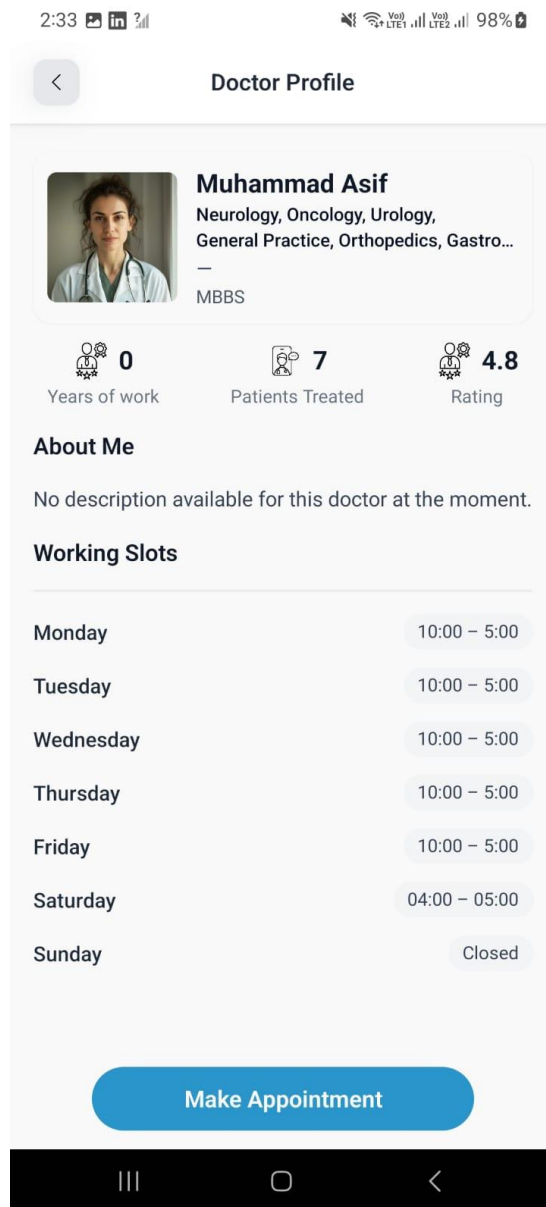


Figure 5.2: Doctor profile

The Home Page of the Doctor Application in Figure 5.3, which provides doctors with quick access to key features such as upcoming appointments, patient requests, and profile management. The interface is designed to be clean and organized, allowing doctors to efficiently manage their daily tasks and navigate the system with ease.

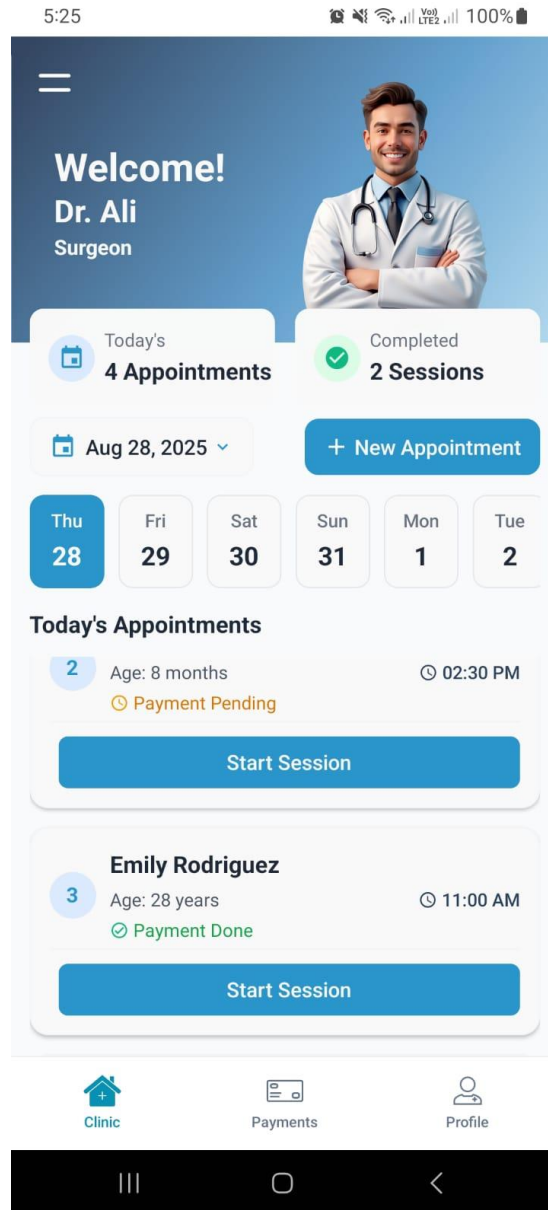


Figure 5.3: Doctor home page

Figure 5.4 shows the Patient Details screen within the Doctor Application, where doctors can view important patient information such as medical history, ongoing treatments, and appointment records. This screen helps ensure informed decision-making by presenting all patient data in a clear and structured format.

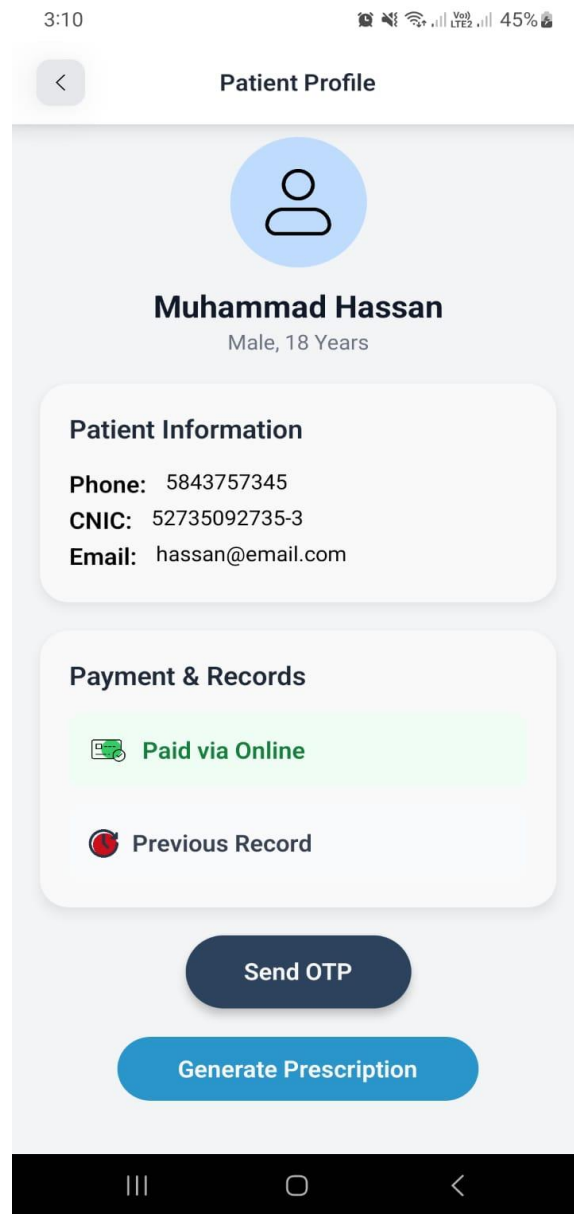


Figure 5.4: Patient Profile

The doctor web page in Figure 5.5 provides an organized interface where doctors can manage appointments, access patient records, and update their professional profile. Its clean layout ensures quick navigation, helping doctors perform tasks efficiently.

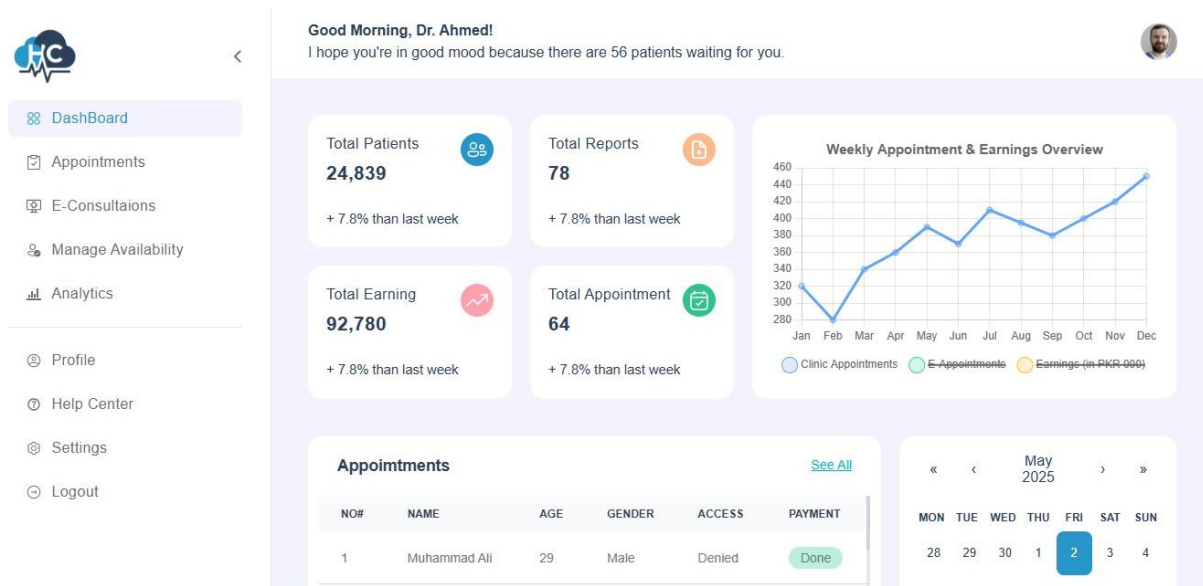


Figure 5.5: Home page web

The Implementation chapter explains how the Health Cloud system was built by transforming the designed architecture into a fully working solution. It describes the core algorithms such as authentication, appointment scheduling, prescription generation, and payment processing that power major system features. The chapter also highlights the external APIs used for payments, notifications, maps, and secure authentication, along with an overview of the user interface design for patients, doctors, and administrators. Overall, this chapter shows how all technical components were integrated to create a secure, efficient, and user-friendly healthcare platform.

Chapter 6

Testing and Evaluation

6 Testing and Evaluation

This chapter demonstrates that the Health Cloud system functions correctly, meets the specified requirements, and provides a reliable and secure experience for users. Testing has been performed using manual approaches, including unit testing, functional testing, and integration testing, to verify that each module works as intended and that the system operates smoothly as a whole. Manual testing helps identify errors that may not be visible during normal use and ensures that all functional requirements are met before deployment.

In addition to manual testing, automated testing has been applied to improve efficiency and accuracy. Automated tools are used to validate key system functionalities, such as login, appointment scheduling, and payment processing, ensuring that they work correctly under different scenarios. By combining manual and automated testing, the Health Cloud system has been thoroughly evaluated for reliability, usability, and performance, confirming its readiness for real-world use.

6.1 Manual Testing

Manual testing is a process where a tester interacts directly with the Health Cloud system to verify that all functionalities are working correctly. Unlike automated testing, manual testing relies on human observation to identify errors, usability issues, and unexpected behavior. It is a critical step in ensuring that the system meets its functional and non-functional requirements and provides a smooth user experience. Manual testing allows testers to simulate real-world usage scenarios, making it easier to detect hidden errors that automated tests may miss.

Manual testing in the Health Cloud system is carried out in multiple stages, including system testing, unit testing, functional testing, and integration testing, before deployment. This ensures that each module works correctly on its own and in combination with other modules, providing a reliable and user-friendly system.

6.1.1 System Testing

System testing is a high-level evaluation conducted after the complete development of the Health Cloud system. Its primary objective is to verify that the system functions as intended and meets all

the specified requirements. Unlike testing individual modules, system testing focuses on the system as a whole, allowing the detection of errors and inconsistencies that might not be evident during the development phase. This ensures that the system operates correctly under real-world conditions.

6.1.1.1 Purpose of System Testing

The purpose of system testing is to confirm that the Health Cloud system meets functional requirements, performs reliably, and is ready for deployment. It helps in detecting hidden errors and inconsistencies in workflows, ensuring that each module interacts correctly with others. System testing also validates that the overall system is stable, secure, and provides a seamless experience for end-users. This stage is critical to guarantee that the software is fully operational before being released for use.

6.1.1.2 Types of System Testing

System testing incorporates several types of evaluation to comprehensively assess the software. Unit testing focuses on individual modules or components, ensuring that they function correctly in isolation. Functional testing evaluates the features of the system to ensure they perform according to the design specifications and meet user requirements. Integration testing checks that multiple modules work together correctly, verifying the flow of data between components and confirming that combined functionalities operate as intended.

6.1.1.3 Importance of Manual System Testing

Manual system testing is performed carefully to validate the complete system. It ensures that the Health Cloud system is reliable, secure, and user-friendly before deployment. By thoroughly testing each module and its interactions, potential errors are detected early, which improves the quality, performance, and overall reliability of the system. This approach provides confidence that the system will function effectively in real-world scenarios.

6.1.2 Unit Testing

Unit testing focuses on verifying the individual modules of the Health Cloud system to ensure that each part functions correctly on its own. It is usually the first step in manual testing and forms the foundation for functional and integration testing.

Unit tests involve as described in Table 6.1 providing specific input values to a module or function and comparing the actual output with the expected result. Any discrepancies are noted and fixed before moving on to higher-level tests.

Table 6.1: Unit Testing Table

Test ID	Module Name	Test Case/Test Script	Description	Input Data	Expected Result	Actual Result	Status
UT01	Login Module	Verify login form	Test login functionality for FYP Committee	Username: L001, Password: 1234	Successfully log in as FYP Committee member	Successfully logged in	Pass
UT02	Profile Module	Edit Profile form	Update user details in profile	Email: test@example.com , Phone: 1234567890	Profile updated successfully	Profile updated successfully	Pass
UT03	Appointment Module	Book Appointment	Patient books an appointment with doctor	Doctor: Dr. A, Date: 25/11/2025, Time: 10:00 AM	Appointment successfully booked	Appointment successfully booked	Pass

UT04	Prescription Module	Create Prescription	Doctor creates a new prescription	Medication: Paracetamol, Dosage: 500mg, Duration: 5 days	Prescription saved in patient record	Prescription saved successfully	Pass
UT05	Payment Module	Process Payment	Process patient consultation payment	Amount: 500, Payment method: Card	Payment processed and receipt generated	Payment processed successfully	Pass
UT06	Notification Module	Send Email Notification	Notify patient of appointment	Email: <u>test@example.com</u> , Message: Appointment Confirmed	Email sent successfully	Email sent successfully	Pass
UT07	Notification Module	Send WhatsApp Notification	Notify patient of prescription update	Phone: 1234567890, Message: Prescription Ready	Message sent successfully	Message sent successfully	Pass

6.1.3 Functional Testing

Functional testing is conducted after unit testing and focuses on verifying that the Health Cloud system's features work according to the specified requirements. The main purpose of functional

testing is to ensure that each function performs correctly in real-world scenarios and meets the design and user expectations.

A common example of functional testing in the Health Cloud system is login with different roles as described in Table 6.2. This test ensures that after login, the correct dashboard is displayed along with the appropriate navigation options for each role, such as patients, doctors, or administrators. Functional testing helps identify issues related to workflows, permissions, and user interface behavior.

Table 6.2: Functional Testing Table

Test ID	Module Name	Test Case/Test Script	Attribute & Value	Expected Result	Actual Result	Status
FT0 1	Login Module	Login as FYP Committee	Username: L001, Password: 1234	Main page with FYP Committee navigation bar	Main page loaded successfully	Pass
FT0 2	Login Module	Login as Patient	Username: P001, Password: 5678	Patient dashboard with appointmen t and profile options	Dashboard loaded successfully	Pass
FT0 3	Login Module	Login as Doctor	Username: D001, Password: abcd	Doctor dashboard with schedule	Dashboard loaded successfully	Pass

				and patient list		
FT04	Appointment Module	Book Appointment	Select doctor, date, and time	Appointment confirmed and notification sent	Appointment booked successfully	Pass
FT05	Prescription Module	Issue Prescription	Enter medication, dosage, duration	Prescription saved in patient record	Prescription saved successfully	Pass

6.1.4 Integration Testing

Integration testing ensures that multiple modules of the Health Cloud system work together correctly and that data flows seamlessly between them. This type of testing is conducted after unit and functional testing and focuses on verifying the interactions between modules to identify errors that might occur when components are combined.

Example integration test scenarios include as described in Table 6.3 logging in and viewing the user profile, uploading student records and verifying database updates, and viewing the list of supervisees. Integration testing helps ensure that combined features operate as intended and that the system provides a smooth user experience.

Table 6.3: Integration Testing Table

Test ID	Modules Involved	Test Case/Test Script	Attribute & Value	Expected Result	Actual Result	Status
---------	------------------	-----------------------	-------------------	-----------------	---------------	--------

IT01	Login & Profile Module	Login as FYP Committee	Username: L001, Password: 1234	Login successful, navigation bar loaded, profile page displayed	Profile page loaded successfully	Pass
IT02	Student Records & Database	Upload Student Record	File: Project1.xlsx	File successfully uploaded, student records updated	Records updated successfully	Pass
IT03	Supervisee Module	View Supervising Students	N/A	List of supervisees displayed correctly	List displayed successfully	Pass
IT04	Appointment & Notification	Book Appointment	Doctor: Dr. A, Date, Time	Appointment booked, notification sent to patient and doctor	Notification sent successfully	Pass

6.2 Automated Testing

Automated testing involves using software tools to test the Health Cloud system instead of performing manual interactions. These tools help verify system functionalities quickly, accurately, and repeatedly. Automated testing is particularly useful for regression testing, performance checks, and verifying repetitive tasks.

The tools used for automated testing as described in Table 6.4 in the Health Cloud project include **Jest** and **Mocha** for backend unit testing, **Postman** for API testing, and **Selenium** for frontend interface testing. Each tool is applied to specific test cases, functional requirements (FR), or non-functional requirements (NFR), and the results are documented to confirm system reliability.

Table 6.4: Automated Testing Table

Tool	Applied On	Test Cases/Requirements Tested	Results
Jest	Backend Modules	Login, Profile, Appointment, Prescription	Pass
Mocha	REST APIs	CRUD operations for users, appointments, payments	Pass
Postman	API Endpoints	Verify responses, status codes, and data integrity	Pass
Selenium	Frontend React Components	Navigation, Forms, Dashboards	Pass

The Testing and Evaluation chapter verifies that the Health Cloud system works correctly, securely, and meets all requirements. Both manual and automated testing were performed, including unit, functional, system, and integration tests, to ensure each module works individually and together. Automated tools like Jest, Mocha, Postman, and Selenium were used to validate core functionalities. Overall, the system was thoroughly tested for performance, reliability, and real-world readiness.

Chapter 7

CONCLUSION

7 Conclusion

The Health Cloud system has been successfully designed and implemented as a comprehensive healthcare management platform. Using the MERN stack (MongoDB, Express.js, React, Node.js), the system provides core functionalities such as patient registration, doctor management, appointment scheduling, prescription management, payment processing, and notifications through email and WhatsApp.

Through careful system design, implementation, and rigorous testing including unit, functional, integration, and automated testing the system has proven to be reliable, secure, and user-friendly. The Health Cloud platform meets the specified requirements and provides a seamless experience for patients, doctors, and administrators. The integration of external APIs, such as payment gateways, Google Maps, and notification services, further enhances the system's usability and efficiency. Overall, the project demonstrates how modern web technologies can be leveraged to improve healthcare management and streamline daily operations in medical institutions.

7.1 Future Work

While the current version of Health Cloud successfully covers core healthcare functionalities, there are several opportunities for future enhancement:

7.1.1 Online Consultation

In future versions, the system can support real-time e-consultations through video or chat, allowing patients to consult doctors remotely [3]. This will increase accessibility and reduce the need for physical visits.

7.1.2 Hospital Deployment

The system can be extended to run across multiple hospitals, allowing centralized management of patients, doctors, and appointments. This will enable hospitals to maintain synchronized records and streamline operations across departments.

7.1.3 Subscription Models

Health Cloud can introduce monthly and yearly subscription plans, providing premium services such as priority appointments, detailed health analytics, or access to exclusive features.

7.1.4 Advanced Analytics and Reporting

Future updates could include dashboards for patients and doctors, generating insights from health records, appointment trends, and prescription histories. Administrators could also receive automated reports on system usage and hospital performance.

7.1.5 Integration with Wearable Devices

Health Cloud could integrate with wearable health monitoring devices, allowing automatic recording of vital signs, activity levels, and other health metrics to improve patient care and preventive health monitoring.

7.1.6 Enhanced Security Features

Future improvements can include multi-factor authentication, encrypted chat for consultations, and AI-driven anomaly detection to ensure patient data privacy and security.

By implementing these enhancements, Health Cloud has the potential to evolve into a fully integrated healthcare management ecosystem, supporting hospitals, doctors, and patients more effectively and providing advanced digital health services.

8 References

- [1] E. Mehraeen, M. Ghazisaeeedi, J. Farzi, and S. Mirshekari, "Security challenges in healthcare cloud computing: A systematic review," *Global Journal of Health Science*, vol. 9, no. 3, pp. 157–175, 2017. (Placeholder1)
- [2] L. Ehwerhemuepha, G. Gasperino, N. Bischoff, et al., "HealtheDataLab – a cloud computing solution for data science and advanced analytics in healthcare with application to predicting multi-center pediatric readmissions," *BMC Medical Informatics and Decision Making*, vol. 20, 115, 2020. [Online]. Available: <https://bmcmmedinformdecismak.biomedcentral.com/articles/10.1186/s12911-020-01153-7> .
- [3] A. Tahir, F. Chen, H. U. Khan, Z. Ming, A. Ahmad, S. Nazir, and M. Shafiq, "A systematic review on cloud storage mechanisms concerning e-healthcare systems," *Sensors*, vol. 20, no. 18, 5392, 2020. [Online]. Available: <https://www.mdpi.com/1424-8220/20/18/5392> [Accessed: 25-Nov-2025].
- [4] H. Karo and G. Karo, "Cloud computing in healthcare: A brief review of advantages and challenges in 2023," *Journal of Advanced Research in Embedded System*, 2023. [Online]. Available: <https://www.thejournalshouse.com> .
- [5] M.-H. Kuo, "Opportunities and challenges of cloud computing to improve health care services," *Journal of Medical Internet Research*, vol. 13, no. 3, e67, 2011. [Online]. Available: <https://www.jmir.org/2011/3/e67/> .
- [6] D. Rizka, H. Hosnya, E.-S. El Horbaty, and A.-B. Salem, "A study on cloud computing architectures for smart healthcare services," *CEUR Workshop Proceedings*, 2023. [Online]. Available: <https://ceur-ws.org/Vol-2753/short12.pdf> .
- [7] P. Khatiwada, H. Bhusal, A. Chatterjee, and M. W. Gerdess, "A proposed access control-based privacy preservation model to share healthcare data in cloud," *arXiv*, 2020. [Online]. Available: <https://arxiv.org/abs/2007.13850> .

[8] J. Romero, P. López, J. L. Vázquez Noguera, C. Cappelletti, D. P. Pinto Roa, and C. Villalba, "Integrated, reliable and cloud-based personal health record: A scoping review," *arXiv*, 2016.
[Online]. Available: <https://arxiv.org/abs/1609.03615>.